CEVAP ANAHTARI

DEVRE TEORİSİ - 2 ARA SINAV 17/11/2016

Ad Soyadı :

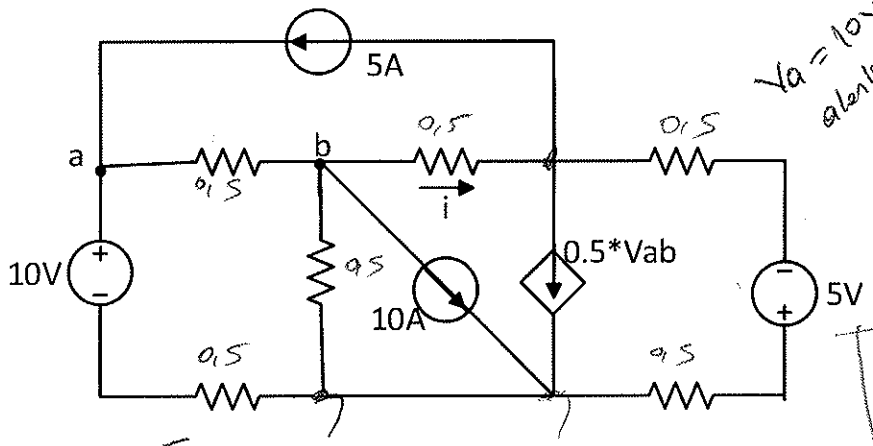
No :

s1	s2	s3	s 4	Toplam

11 01

Kremele sözümü dinle! Son soruda son 3 dakika kurtarır

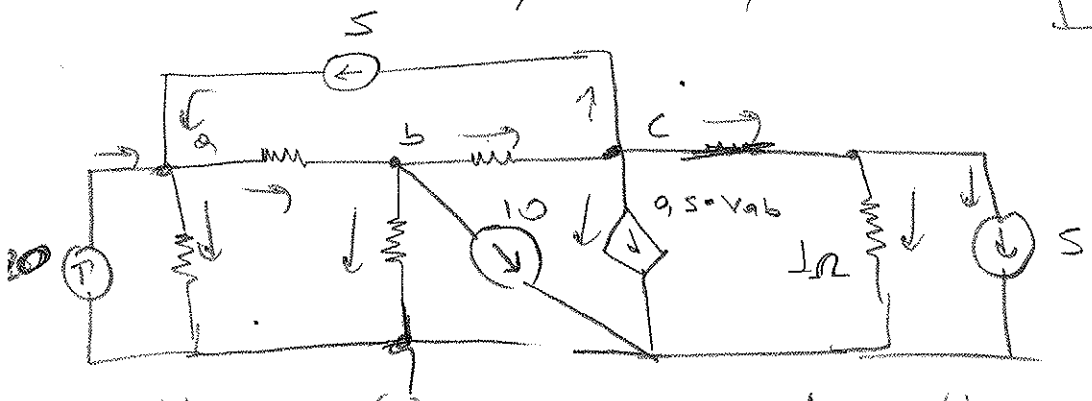
Soru 1: Aşağıda verilen devrede ***i*** akımını **Düğüm Gerilimleri Yöntemi** ile hesaplayınız. (Tüm dirençler 0.5 Ω 'dur.)



Va = 10V alırsak 4 puan kurtarır

son ekrat toprak kısmak gerek.

$i = \frac{V_b - V_c}{0,5} = \frac{55}{7} = 7,8571$ A



10/0,5 = 20 A

5/1 = 5 A

a noktası için (6)

$\frac{V_a}{0,5} + \frac{V_a - V_b}{0,5} - 20 - 5 = 0$

$4V_a - 2V_b = 25$ (1)

b noktası için (6)

$-\frac{(V_a - V_b)}{0,5} + \frac{V_b}{0,5} + 10 + \frac{V_b - V_c}{0,5} = 0$

3 denklemle 3 bilinmeyen dediğim için sonuçta yanlış işlem bulursa 2 puan kırarım

$V_a - 3V_b + V_c = 5$ (2)

ilk adımda denklemi sonuç için doğru ise puan kırmıyorum

c noktası için (6)

$-\frac{(V_b - V_c)}{0,5} + 5 + 5 + \frac{V_c}{1} + 0,5 \cdot (V_a - V_b)$

$V_a - 5V_b + 6V_c = -20$ (3)

$\begin{bmatrix} 4 & -2 & 0 \\ 1 & -3 & 1 \\ 1 & -5 & 6 \end{bmatrix} \begin{bmatrix} V_a \\ V_b \\ V_c \end{bmatrix} = \begin{bmatrix} 25 \\ 5 \\ -20 \end{bmatrix}$

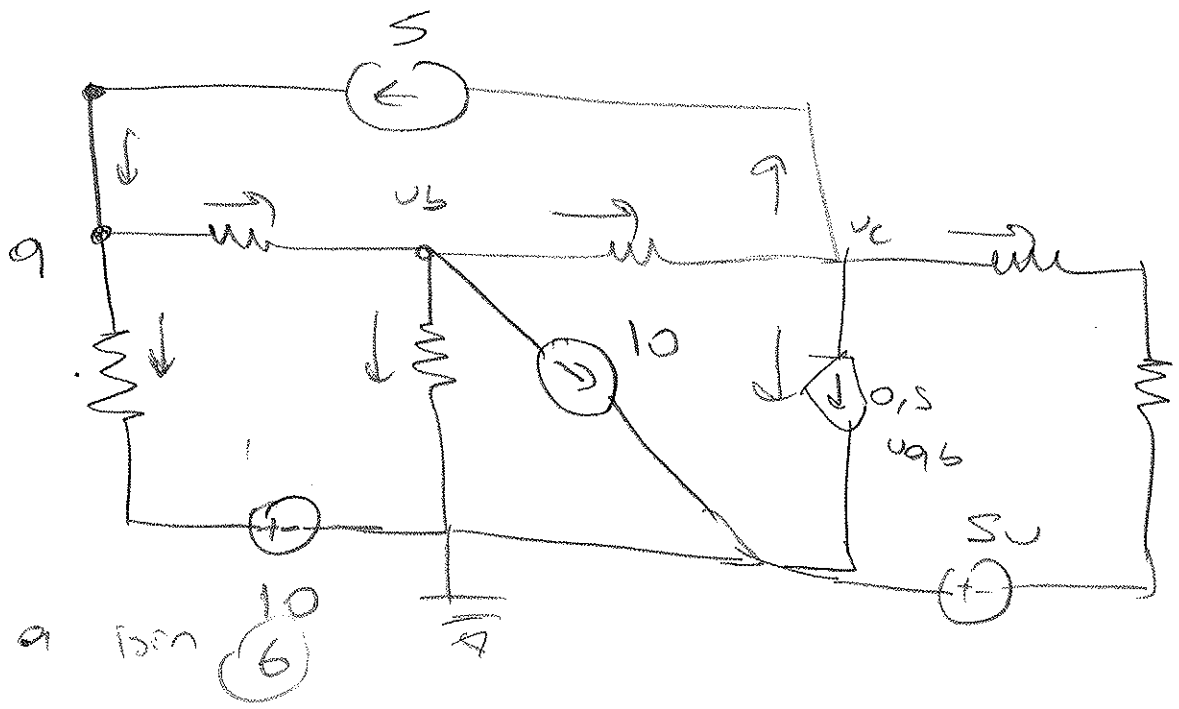
$V_b = \frac{\begin{vmatrix} 4 & 25 & 0 \\ 1 & 5 & 1 \\ 1 & -20 & 6 \end{vmatrix}}{\begin{vmatrix} 4 & -2 & 0 \\ 1 & -3 & 1 \\ 1 & -5 & 6 \end{vmatrix}} = \frac{-25}{14}$

$V_b = -1,7857$ (2)

$V_a = \frac{75}{14} = 5,3571$ (2)

$V_c = -\frac{80}{9} = -5,7143$ (2)

İşlem kabul 2 puan kırılır



a צומת (6)

$$\frac{V_a - 10}{0,5} + \frac{V_a - V_b}{0,5} - 5 = 0 \qquad 4V_a - 2V_b = 25$$

b צומת (6)

$$-\frac{(V_a - V_b)}{0,5} + \frac{V_b}{0,5} + 10 + \frac{V_b - V_c}{0,5} = 0 \qquad V_a - 3V_b + V_c = 5$$

c צומת (6)

$$\frac{(V_b - V_c)}{0,5} + 0,5(V_a - V_b) + 5 + \frac{V_c + 5}{1} = 0 \qquad V_a - 5V_b + 6V_c = -20$$

$V_a = (2)\ \frac{75}{14}$

$V_b = (2)\ -\frac{25}{14}$

$V_c = (2)\ -\frac{40}{7}$

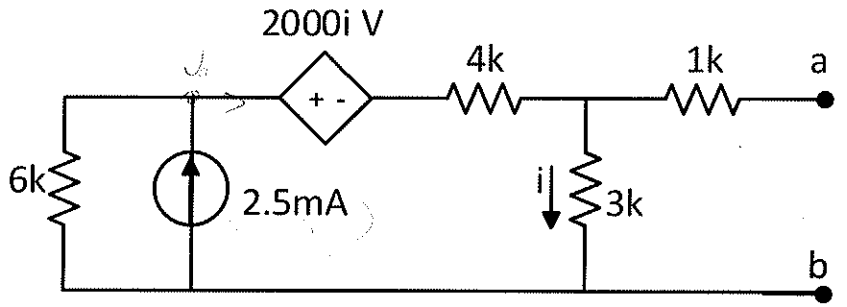
$i = 1 \quad \frac{55}{7}$

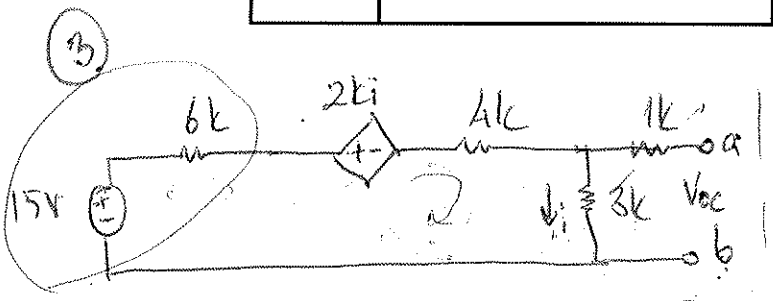
DEVRE TEORİSİ - 2 ARA SINAV 17/11/2016

Ad Soyadı :

No :

Soru 2: Verilen devrenin a-b uçları arası Thevenin Eşdeğer Devresinin elde ediniz. a-b uçlarından çekilebilecek azami güç değerlerini belirleyiniz.



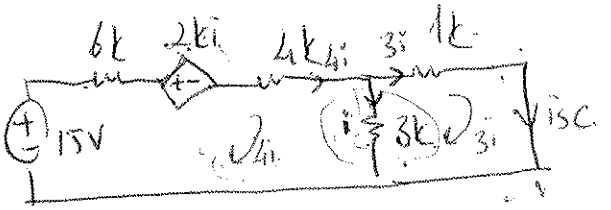


$-15 + 6ki + 2ki + 4ki + 3ki = 0$ (KVL)

$15ki = 15$

$i = 1mA$

$V_{oc} = 3k \cdot i = 3V$ (2)



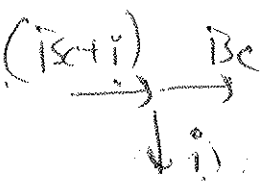
$-15 + 4i \cdot 6k + 2ki + 4k \cdot 4i + 3k \cdot i = 0$ (5)

$45ki = 15$

$i = \frac{1}{3} mA$

$I_{sc} = 3 \cdot i = 1mA$ (2)

$R_{th} = \frac{V_{oc}}{I_{sc}} = \frac{3V}{1mA} = 3k\Omega$ (3)



$R_{th} = R_{L\,max} = 3k\Omega$

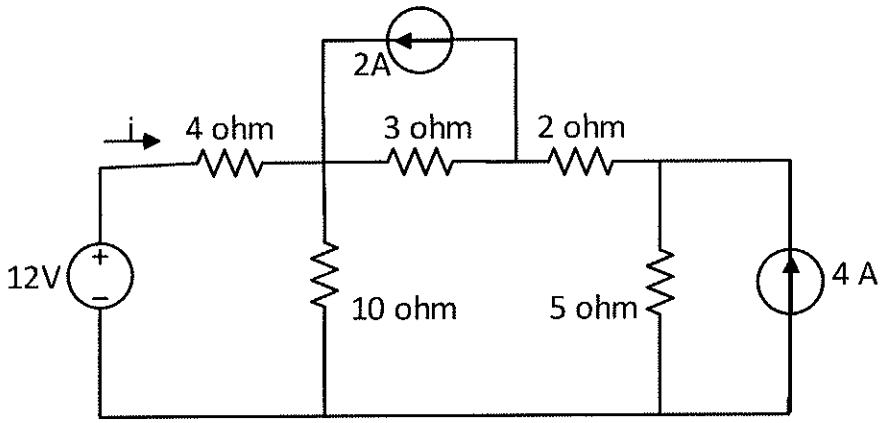
$P_{L\,max} = \frac{V_{oc}^2}{4 \cdot R_{th}} = \frac{3^2}{4 \cdot 3k} = 0{,}75\,mW$ (5)

DEVRE TEORİSİ - 2 ARA SINAV 17/11/2016

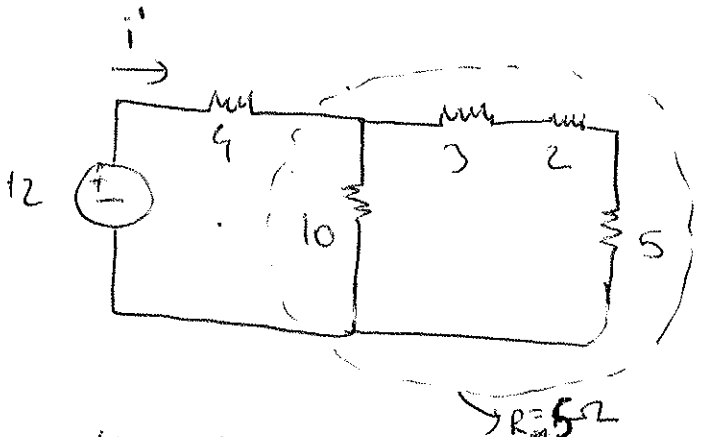
Ad Soyadı :

No :

Soru 3: i akımını süper pozisyon teoremi ile belirleyiniz. (Kaynak dönüşümü yapmayınız.)



12 V'luk kaynak devrede

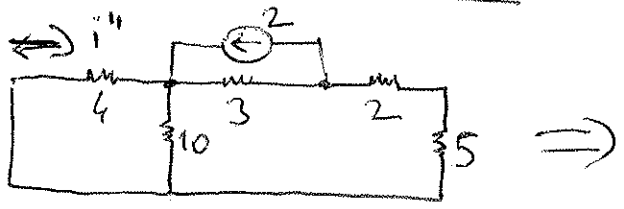


$$i' = \frac{12}{4+5} = \frac{4}{3}\ A$$

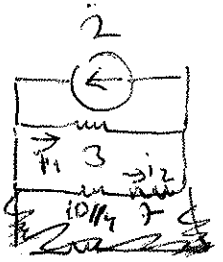
(6)

$R_{eq} = 5\Omega$

2 A'lik kaynak devrede



(9)



~~$R_{eq} = 3 // ((10//4)+7)$~~

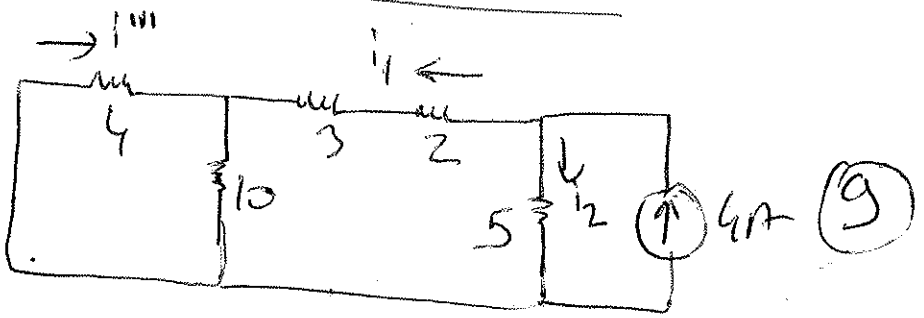
$$i_2 = \frac{3}{3+(10//4)+7} \cdot 2$$

$i_2 = 0{,}467$

$$i'' = -i_2 \cdot \frac{10}{10+4} = \frac{-1}{3}\ A$$

$$i'' = -\frac{1}{3}\ A$$

4 A'lık kaynak devrede



$$i_1 = 4A \cdot \frac{5}{5+(5+4//10)} = 1,556 = \frac{14}{9} A$$

$$i''' = -i_1 \cdot \frac{10}{10+4} = \frac{-14}{9} \cdot \frac{10}{14} = -\frac{10}{9} A$$

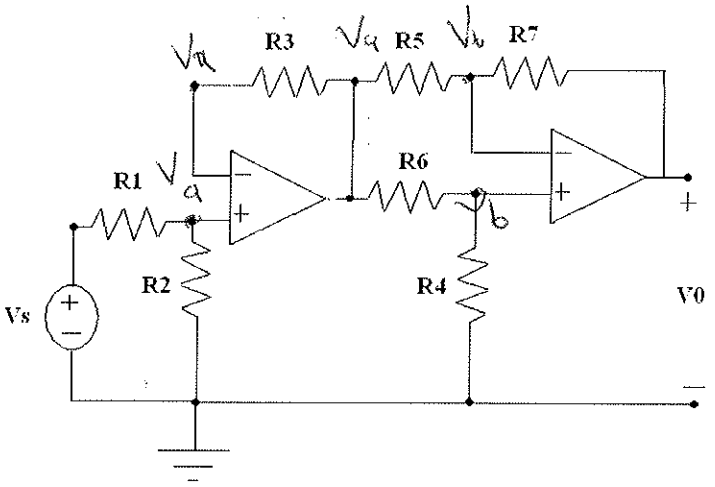
$$i = i' + i'' + i''' = \frac{4}{3} - \frac{1}{3} - \frac{10}{9} = -\frac{1}{9} A \quad (1)$$

DEVRE TEORİSİ - 2 ARA SINAV 17/11/2016

Ad Soyadı :

No :

Soru 4: Opamplı devrede V_0 çıkış gerilimini bulunuz. (R1=R2=1Ω, R3=R4=2Ω, R5=R6=R7=3Ω, Vs=12V)



R_3 üzerinden akım akmaz. Onun için R_3 üzerindeki potansiyel fark 0'dır yani iki uç gerilimi birbirine eşittir.

(1) $\frac{V_a - V_s}{R_1} + \frac{V_a}{R_2} = 0 \Rightarrow \frac{V_a - V_s}{1} + \frac{V_a}{1} = 0$

(2) $2V_a = V_s \qquad V_a = \frac{V_s}{2} = \frac{12}{2} = 6\ V$

(2) $\frac{V_b - V_a}{R_6} + \frac{V_b}{R_4} = 0 \Rightarrow \frac{V_b - V_a}{3} + \frac{V_b}{2} = 0$ (5)

(2) (3)

$5V_b = 2V_a \qquad V_b = \frac{2V_a}{5} = 2,4\ V$ (3)

(3) $\frac{V_b - V_a}{3} + \frac{V_b - V_0}{3} = 0 \Rightarrow 2V_b - V_a - V_0 = 0$

$V_0 = 2V_b - V_a = 2 \cdot 2,4 - 6 = -1,2\ V$

$\boxed{V_0 = -1,2\ V}$

DEVRE TEORİSİ - 2 ARA SINAV 17/11/2016

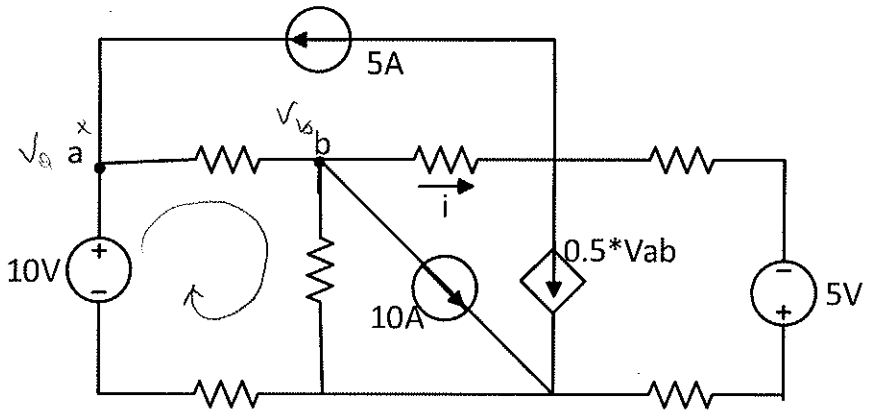
Ad Soyadı : İrfan Civil

No : 141024002

s1	s2	s3	s 4	Toplam
0	0	0	0	0

Nodal voltaj ile

Soru 1: Aşağıda verilen devrede ***i*** akımını **Düğüm Gerilimleri Yöntemi** ile hesaplayınız. (Tüm dirençler 0.5 Ω 'dur.)



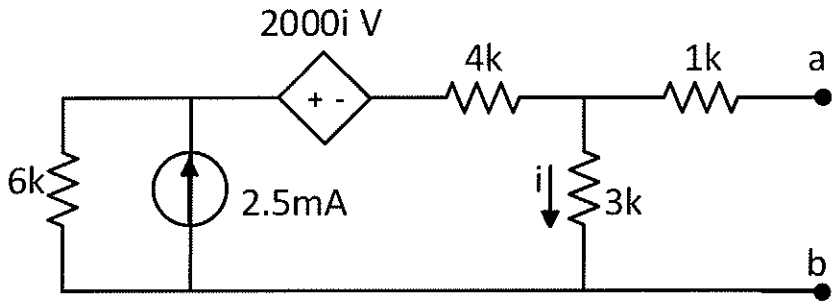
$(V_a - V_b) = i\,(0.5)$

DEVRE TEORİSİ - 2 ARA SINAV 17/11/2016

Ad Soyadı : İrfan Civili

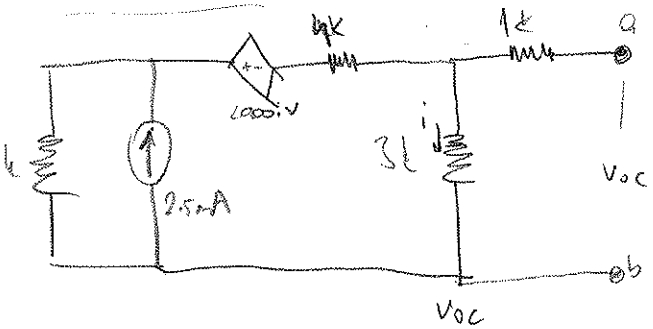
No : 141024002

Soru 2: Verilen devrenin a-b uçları arası Thevenin Eşdeğer Devresinin elde ediniz. a-b uçlarından çekilebilecek azami güç değerlerini belirleyiniz.

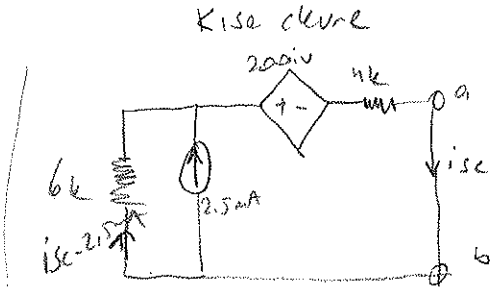


$$\left(Güç = \frac{V_{oc}^2}{R_{TH}}\right)$$

Açık devre



Kısa devre



$$\frac{2000i - V_{oc}}{6k} - 2{,}5mA + \frac{V_{oc}}{1k} + \frac{2000i}{4k} + \frac{V_{oc}}{3k} = 0$$

$$\frac{2000i - V_{oc}}{6k} + \frac{6V_{oc}}{6k} + \frac{2V_{oc}}{6k} - 2{,}5mA = 0$$

$$\frac{2000i - 7V_{oc}}{6} = 2{,}5mA$$

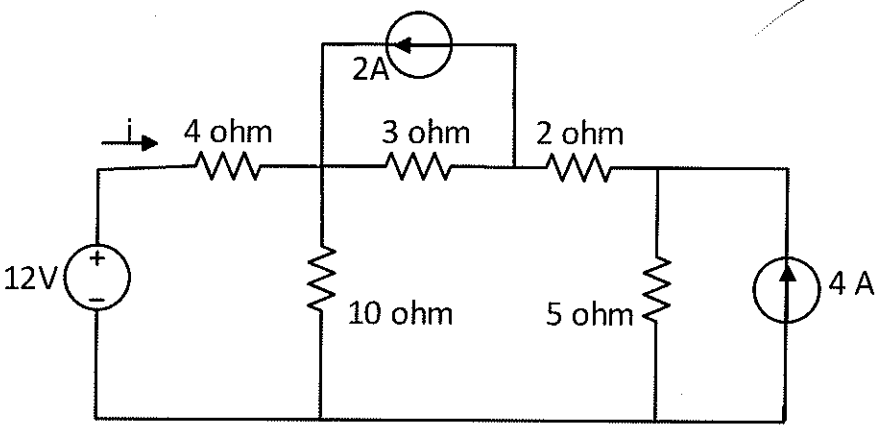
$$V_{oc} = 283{,}57i$$

DEVRE TEORİSİ - 2 ARA SINAV 17/11/2016

Ad Soyadı : İrfan Civil

No : 1411621002

Soru 3: i akımını süper pozisyon teoremi ile belirleyiniz. (Kaynak dönüşümü yapmayınız.)

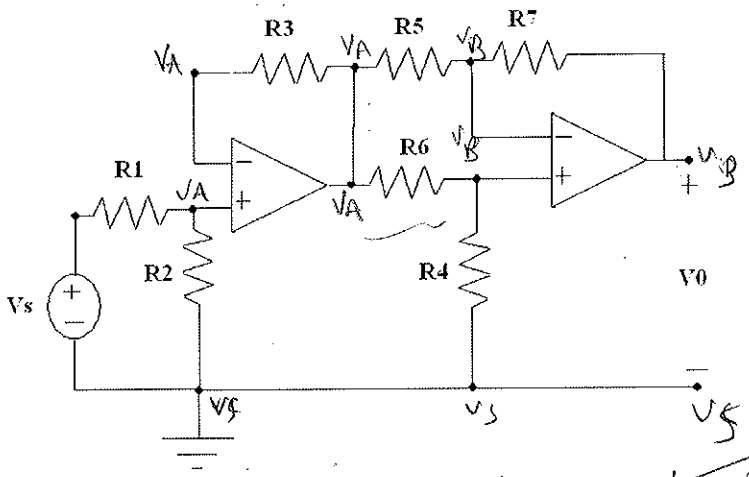


DEVRE TEORİSİ - 2 ARA SINAV 17/11/2016

Ad Soyadı : İrfan Civil

No : 14102403 2

Soru 4: Opamplı devrede V_0 çıkış gerilimini bulunuz. (R1=R2=1Ω, R3=R4=2Ω, R5=R6=R7=3Ω, Vs=12V)



$V_0 = V_B - V_S$

$\frac{V_A - V_B}{R_6} = \frac{V_A - V_B}{3\Omega} = ?$

$\frac{V_A - V_B}{3} = \frac{V_A - 12}{1}$

$V_A - V_B = 3V_A - 36$

$36 - V_B = 2V_A$

$2V_A + V_B = 36$

$\frac{V_A - 12}{R_1} = \frac{V_A - 12}{1\Omega} = ?$

$2V_A - V_B = -12$

$V_A = 12$

$V_B = 12$

$\frac{V_A - V_S}{R_2 = 1} = \frac{V_B - V_S}{R_4 = 2}$

$2V_A - 24 = V_B - 12$

$2V_A - V_B = -12$

$V_0 = 0$

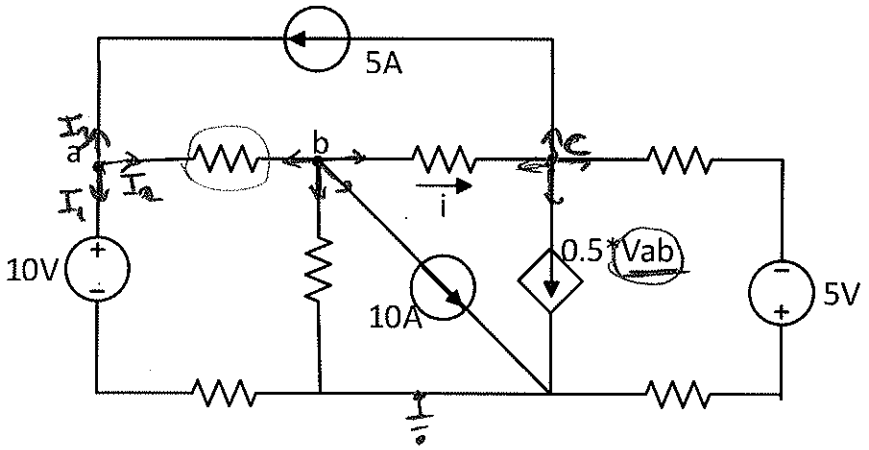
DEVRE TEORİSİ - 2 ARA SINAV 17/11/2016

Ad Soyadı : Müjdat Gödek

No : 131024080

s1	s2	s3	s 4	Toplam
23	25	25	25	98

Soru 1: Aşağıda verilen devrede ***i*** akımını **Düğüm Gerilimleri Yöntemi** ile hesaplayınız. (Tüm dirençler 0.5 Ω 'dur.)



a düğümü için

$I_1 + I_2 + I_3 = 0$

$\frac{V_a + 10}{0,5} + \frac{V_a - V_b}{0,5} - 5 = 0$

$\frac{V_a + 10 + V_a - V_b}{0,5} = 5$

$V_a + 10 + V_a - V_b = 2,5$

$2V_a - V_b = 2,5 - 10$

$\boxed{2V_a - V_b = -7,5}$ 1. denklem (4)

b düğümü için

$\frac{V_b - V_a}{0,5} + \frac{V_b}{0,5} + \frac{V_b - V_c}{0,5} + 10 = 0$

$\frac{V_b - V_a}{0,5} + \frac{V_b}{0,5} + \frac{V_b - V_c}{0,5} = -10$

$V_b - V_a + V_b + V_b - V_c = -5$

$\boxed{-V_a + 3V_b - V_c = -5}$ 2 denklem (6)

c düğümü için

$5 + \frac{V_c - V_b}{0,5} + 0,5.(0,5.\frac{V_a - V_b}{0,5}) + \frac{V_c + 5}{0,5 + 0,5} = 0$

$5 + \frac{V_c - V_b}{0,5} + \frac{0,125(V_a - V_b)}{0,5} + \frac{V_c + 5}{1} = 0$

$\frac{5}{1} + \frac{V_c - V_b}{0,5} + \frac{0,125.V_a - 0,125V_b}{0,5} + \frac{V_c + 5}{1} = 0$

$2V_c - 2V_b + 0,5V_a - 0,5V_b + V_c = -10$

$\boxed{0,5V_a - 2,5V_b + 3V_c = -10}$ 3. denklem (6)

$$\begin{bmatrix} 2 & -1 & 0 & | & -7,5 \\ -1 & 3 & -1 & | & -5 \\ 0,5 & -2,5 & 3 & | & -10 \end{bmatrix} = \begin{bmatrix} V_a \\ V_b \\ V_c \end{bmatrix}$$

$\det \Delta = 10,5$

$$\left[\begin{array}{ccc|c} 2 & -1 & 0 & -7{,}5 \\ -1 & 3 & -1 & -5 \\ 0{,}5 & -2{,}5 & 3 & -10 \end{array}\right] = \begin{bmatrix} V_a \\ V_b \\ V_c \end{bmatrix}$$

$$\det \Delta = 10{,}5$$

$$V_b = \frac{\det |V_b|}{\det \Delta} = \frac{\begin{vmatrix} 2 & -7{,}5 & 0 \\ -1 & -5 & -1 \\ 0{,}5 & -10 & 3 \end{vmatrix}}{\begin{vmatrix} 2 & -1 & 0 \\ -1 & 3 & -1 \\ 0{,}5 & -2{,}5 & 3 \end{vmatrix}} = \frac{-68{,}75}{10{,}5} = -6{,}547 \text{ Volt}$$

$$V_c = \frac{\det |V_c|}{\det \Delta} = \frac{\begin{vmatrix} 2 & -1 & -7{,}5 \\ -1 & 3 & -5 \\ 0{,}5 & -2{,}5 & -10 \end{vmatrix}}{10{,}5} = \frac{-80}{10{,}5} = -7{,}619 \text{ Volt}$$

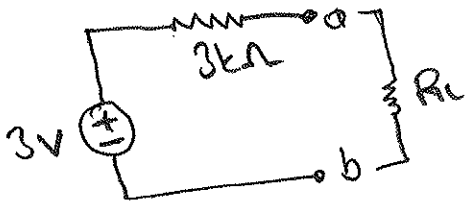
$$i = \frac{V_b - V_c}{0{,}5} = \frac{-6{,}547 - (-7{,}619)}{0{,}5} = \frac{7{,}619 - 6{,}547}{0{,}5}$$

$$\boxed{i = 2{,}144 \text{ A}}$$

$V_{oc} = V_{th} = V_{ab} = 3V$

$I_{sc} = 1mA$

$$R_{th} = \frac{V_{th}}{I_{sc}} = \frac{3}{1mA} = 3k\Omega$$ (m)



Azami güç için

$R_{th} = R_L$

$R_L = 3k\Omega$

$$P = \frac{V_{oc}^2}{4R_L} = \frac{3^2}{4.3000} = 7{,}5 \times 10^{-4} W = 0{,}75 \times 10^{-3} W$$ (n)

$= 0{,}75\ mW$

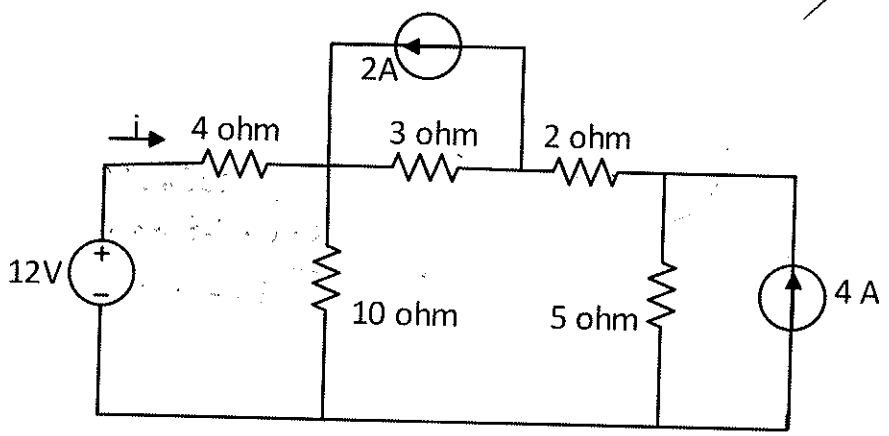
Ad Soyadı : Müjdat Güdek

No : 131024080

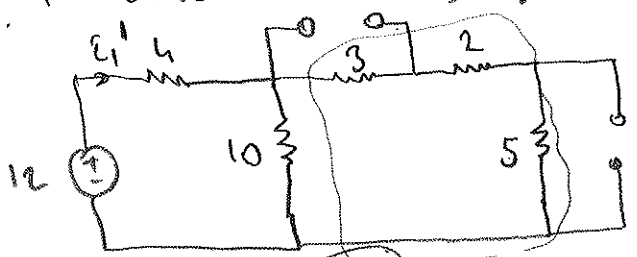
DEVRE TEORİSİ - 2 ARA SINAV 17/11/2016

25

Soru 3: i akımını süper pozisyon teoremi ile belirleyiniz. (Kaynak dönüşümü yapmayınız.)



12 Voltluk Gerilim kaynağı devrede



$3+2+5=10\,\Omega$



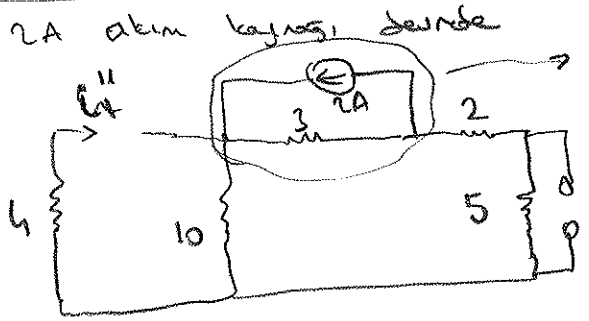
$\frac{10 \cdot 10}{10+10} = \frac{100}{20} = 5\,\Omega$



~~Req = 4+5 = 9Ω~~

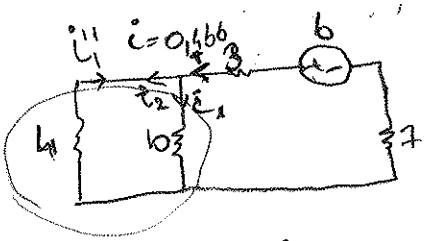
$i_1' = \frac{12}{9}\ A$

2A akım kaynağı devrede

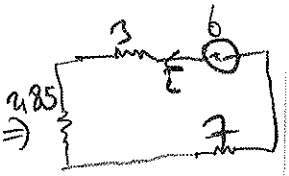


→ Kaynak dönüşümü

$V = 2 \cdot 3 = 6V \quad R = 3\,\Omega$



$\frac{4 \cdot 10}{10+4} = 2{,}85\,\Omega$

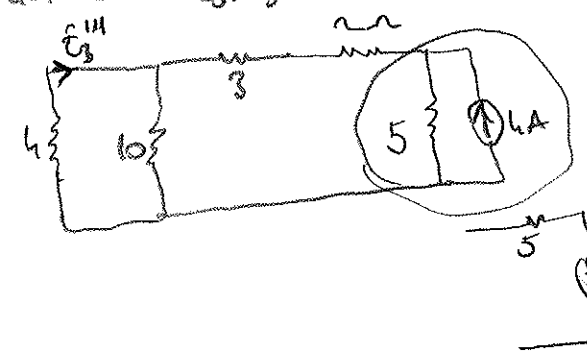


$R_{eş} = 12{,}85$

$i = \frac{6}{12{,}85}$

$i = 0{,}466\ A$

4A akım kaynağı devrede



$V = 4 \cdot 5 = 20V$

$R = 5$

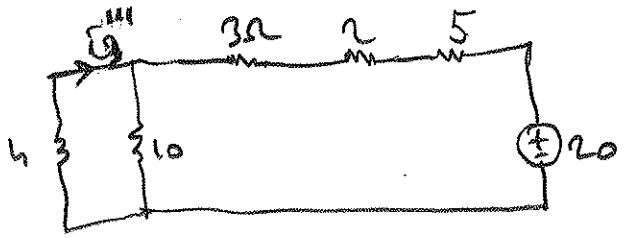
9

$i_2 = i \cdot \frac{10}{14}$

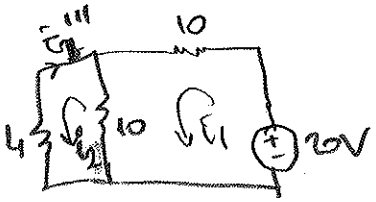
$i = \frac{0{,}466 \cdot 10}{14} = +0{,}332\ A$

$i_2 = -i_1'' = -0{,}332$

4A akım kaynağı devrede



$\Rightarrow$



$-20 + 10i_1 + 10i_1 - 10i_2 = 0$

$4i_2 + 10i_2 - 10i_1 = 0$

$20i_1 - 10i_2 = 20$ 1. denk

$2/ -10i_1 - 14i_2 = 0$

$20i_1 - 10i_2 = 20$

$-2i_1 + 28i_2 = 0$

$+18i_2 = 20$

$i_2 = \frac{20}{18} = 1,11\ A$

$i_1''' = -i_2 = -1,11\ A$

9

$i = i_1' + i_1'' + i_1''' =$

$i = \frac{12}{9} - 0,332 - 1,11$

$i = -0,108\ A = -108\ mA$

1

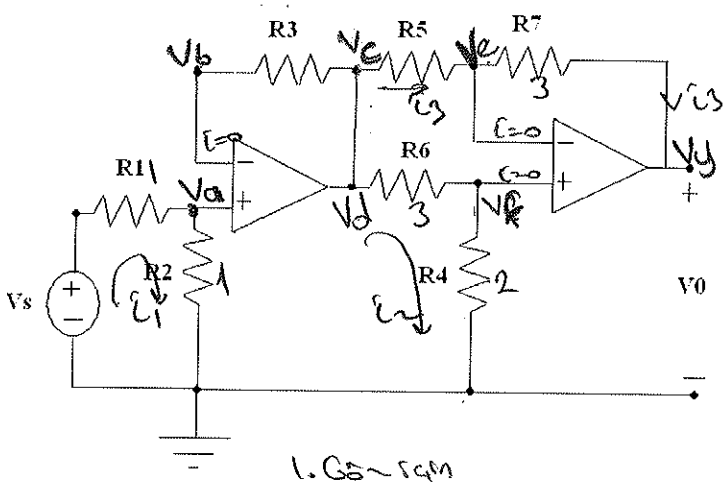
DEVRE TEORİSİ - 2 ARA SINAV 17/11/2016

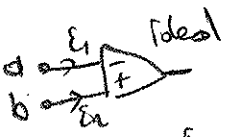
Ad Soyadı : Miğdat Gödek

No : 131024080

25

Soru 4: Opamplı devrede V_0 çıkış gerilimini bulunuz. (R1=R2=1Ω, R3=R4=2Ω, R5=R6=R7=3Ω, Vs=12V)





$\varepsilon_1 = \varepsilon_2 = 0$

$V_a = V_b$

1. Göz için

$-12 + 1.\varepsilon_1 + 1.\varepsilon_1 = 0$

$2\varepsilon_1 = 12$

$\varepsilon_1 = 6A$

$V_0 = \varepsilon_1 \cdot R_2 = 6 \cdot 1 = 6V$ ✓

$V_a = V_b = 6V$ ✓ (6)

V_b den akım aktığı için $i = 0$

$V_b = V_c$ olur $V_c = 6V$ (3)

Aynı noktada $V_c = V_d = 6V$

$\varepsilon_2 = \frac{V_d}{3+2} = \frac{6}{5} = 1{,}2\,A$ (8)

$V_f = \varepsilon_2 \cdot R_4 = 1{,}2 \cdot 2 = 2{,}4\,V$

$V_f = V_e = 2{,}4\,V$

$\frac{V_c - V_e}{R_5} = \varepsilon_3$ olsun

$\frac{6 - 2{,}4}{3} = \varepsilon_3 = 1{,}2\,A$

$\varepsilon_3 = \frac{V_e - V_y}{R_7} = \frac{2{,}4 - V_y}{3}$

$1{,}2 = \frac{2{,}4 - V_y}{3}$ (7)

$3{,}6 = 2{,}4 - V_y$

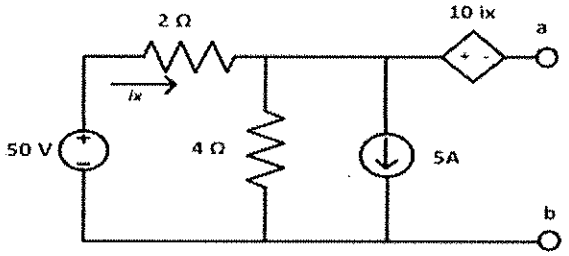
$V_y = 2{,}4 - 3{,}6$

$V_y = -1{,}2$ Volt $= V_0 \Rightarrow V_0 = -1{,}2\,V$

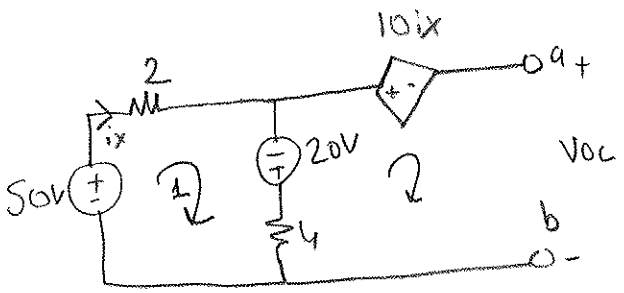
DEVRE TEORİSİ - 1 Final 10/01/2017

Ad Soyadı : Bülent Kepeli
No : 141024076

s1	s2	s3	s 4	Toplam
25	25	25	25	100



Soru 1:
Verilen devrenin a-b uçları arası Thevenin Eşdeğer Devresinin elde ediniz. a-b uçlarından çekilebilecek azami güç değerini belirleyiniz.



1.Çevre

$-50 + 2ix - 20 + 4ix = 0$

$6ix = 70$

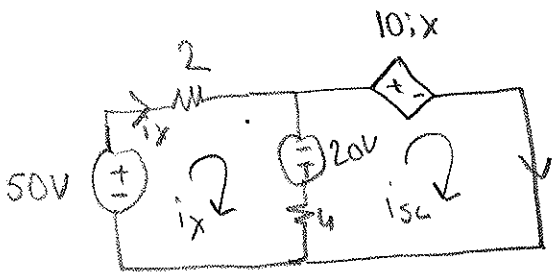
$ix = 70/6$ A

2.Çevre

$10ix + Voc - 4ix + 20 = 0$

$Voc = -6ix - 20$

$Voc = -90V$ (9)



I/ $-50 + 2ix - 20V + 4ix - 4isc = 0$

$6ix - 4isc = 70$

II/ $10ix + 4isc - 4ix + 20 = 0$

$6ix + 4isc = -20$

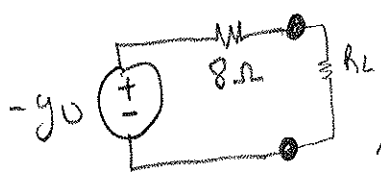
I/ $6ix - 4isc = 70$

$6ix + 4isc = -20$

$12ix = 50$

$ix = 4,1667$ A $\Rightarrow$ $isc = -11.25$ A (9)

$\Rightarrow R_{TH} = \frac{Voc}{isc} = \frac{-90}{-11.25} = 8\Omega$ (3)



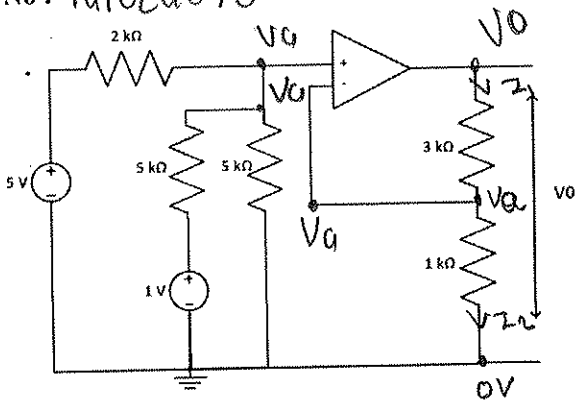
(4)

$P_{max} = \frac{V_s^2}{4R_{TH}} = \frac{(-90)^2}{4.8} = 253,125$ watt //

25

Ad Soyadı: Bulent Kupeli

No: 161026076



Soru 2: Vo gerilimini bulunuz.

$$\frac{Va-5}{2k} + \frac{Va-1}{5k} + \frac{Va}{5k} = 0$$

$$Va\left(\frac{1}{2k} + \frac{1}{5k} + \frac{1}{5k}\right) = \frac{5}{2k} + \frac{1}{5k}$$

$$Va \cdot 9\times10^{-4} = 2.7\times10^{-3}$$

$$Va = 3V$$ ✓

Vo için;

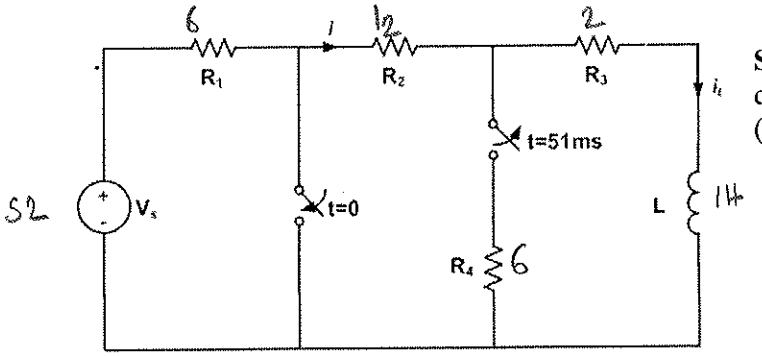
$$\frac{Va-Vo}{3k} + \frac{Va-0}{1k} = 0$$

$$\frac{3}{3k} + \frac{3}{1k} = \frac{Vo}{3k} \Rightarrow Vo = 12V$$ ✓

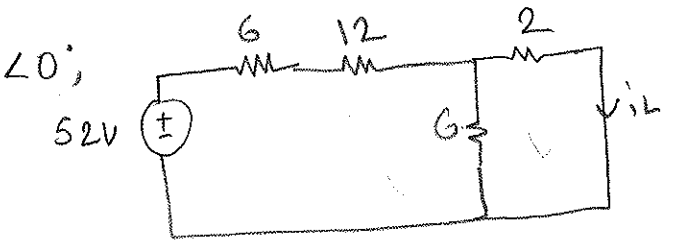
Ad Soyadı : Bulut Kupeli

No : 141024076

25



Soru 3: t>0 için $i_L(t)$'yi hesaplayınız. t=0⁻ anında devre kararlı haldedir.
(R1=6Ω, R2=12Ω, R3=2Ω, R4=6Ω, L=1H,Vs=52V)

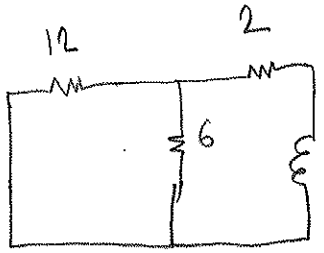


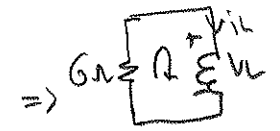
$R_{eq} = \frac{2.6}{8} + 18 = 19.5\,\Omega$

$i = \frac{52}{19.5} = 2.67\,A$

$i_L(0^-) = \frac{2.67 \times 6}{8} = 2A$

3



$\Rightarrow$ 

$6i_L + V_L = 0$

$L\frac{di_L}{dt} + 6i_L = 0 \Rightarrow \frac{di_L}{dt} + \frac{6}{L} i_L = 0$

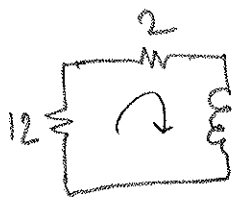
19

$S+6=0 \Rightarrow S_1 = -6$

$i_n = A e^{-6t}$, $i_f = 0$ $\Rightarrow i_L(t) = A e^{-6t} \Rightarrow i_L(0^-) = 2 = A \cdot 1$, $A = 2$

$i_L(t) = 2e^{-6t}$ $0 < t < 51ms$

$<51ms$; $i_L(t) = 2e^{-6t} = 2e^{-6 \times 51 \times 10^{-3}} = 1,4727\,A$ 2



$>51ms$;

$1H i_L + V_L = 0$ 5

$L\frac{di_L}{dt} + 1H i_L = 0$

$S+1H=0$, $S_1 = -1H$ $\Rightarrow i_L(t) = A e^{-1Ht}$ 2

$1,4727 = A \cdot e^{0}$

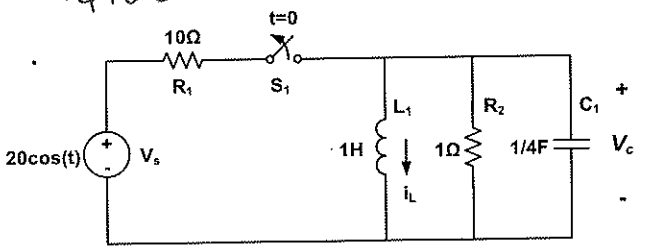
$i_L(t) = 1,47\,A^{-1H(t-51\times10^{-3})}$ $t > 51ms$

$$i_L(t) = \begin{cases} i_L(t) = 2e^{-6t}, & 0 < t < 51ms \\ i_L(t) = 1,47 A^{-1H(t-51\times10^{-3})}, & t > 51ms \end{cases}$$

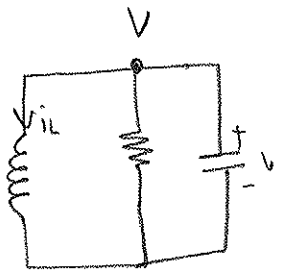
2

Ad Soyadı : Bülent Küpeli

No : 141026076



Soru 4:
t>0 için $V_c(t)$'yi bulunuz. **t=0**$^-$ anında devre kararlı haldedir. ($V_c(0)=2.5V$, $i_L(0)=-1A$)



$$\frac{V}{1} + i_L + i_C = 0 \qquad V = L\frac{di_L}{dt}$$

$$i_L = -C\frac{dV_c}{dt} - V \Rightarrow$$

$$V = L\left(-C\frac{d^2V}{dt^2} - \frac{dV}{dt}\right) \Rightarrow LC\frac{d^2V}{dt^2} + L\frac{dV}{dt} + V = 0 \Rightarrow$$

$$\frac{d^2V}{dt^2} + \frac{1}{C}\cdot\frac{dV}{dt} + \frac{V}{LC} = 0 \Rightarrow \frac{d^2V}{dt^2} + 4\frac{dV}{dt} + 4V = 0 \Rightarrow s^2+4s+4=0$$

$s^2+4s+4=0$

$\Delta = b^2-4ac = 16-16 = 0$

$$s_1 = \frac{-4+0}{2} = -2 \qquad s_2 = \frac{-4-0}{2} = -2 \qquad \Rightarrow s_1 = s_2 = -2$$

$$n = e^{-2t}(A+B.t) \qquad f = 0$$

$$V_c(t) = Ae^{-2t} + B.t.e^{-2t}$$

$$V_c(0) = 2.5 = A + 0 \Rightarrow A = 2.5$$

$$V_c(t) = 2.5e^{-2t} + B.t.e^{-2t}$$

$$\frac{dV_c}{dt} = -\frac{V+i_L}{C} \text{ (devredeki denklemden)} \Rightarrow \left.\frac{dV_c}{dt}\right|_{t=0} = -\left|\frac{2.5-1}{\frac{1}{4}}\right| = -6$$

$$\left.\frac{dV_c}{dt}\right|_{t=0} = -6 = 2.5\times(-2).e^{-2.0} + B.e^{-2.0} - 2.B.t.e^{-2.0}$$

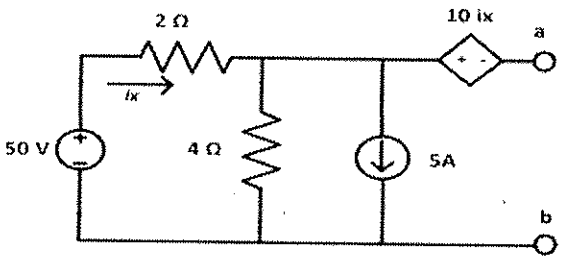
$$B = -1$$

$$V_c(t) = 2.5e^{-2t} - t.e^{-2t}$$

DEVRE TEORİSİ - 1 Final 10/01/2017

Ad Soyadı : Baran Murt
No : 141024024

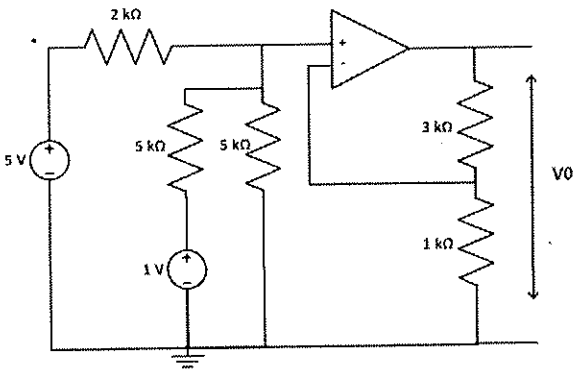
s1	s2	s3	s 4	Toplam
0	0	0	0	0



Soru 1:
Verilen devrenin a-b uçları arası Thevenin Eşdeğer Devresinin elde ediniz. a-b uçlarından çekilebilecek azami güç değerini belirleyiniz.

Ad Soyadı :

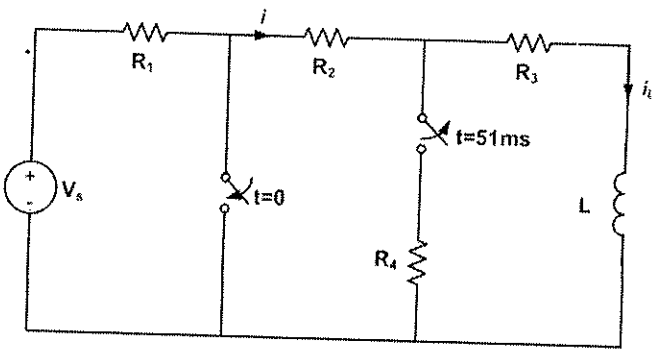
No :



Soru 2: Vo gerilimini bulunuz.

Ad Soyadı :

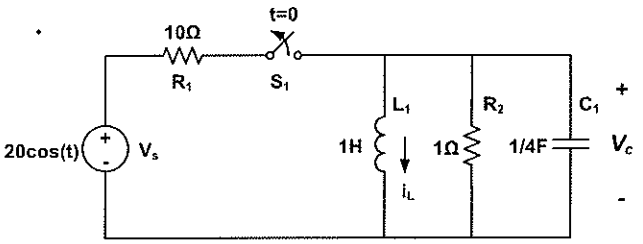
No :



Soru 3: $t>0$ için $i_L(t)$'yi hesaplayınız. $t=0^-$ anında devre kararlı haldedir.
(R1=6Ω, R2=12Ω, R3=2Ω, R4=6Ω, L=1H,Vs=52V)

Ad Soyadı :

No :



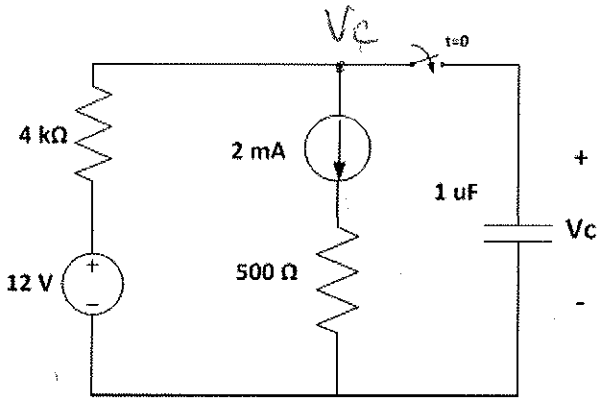
Soru 4:
t>0 için **$V_c(t)$**'yi bulunuz. **$t=0^-$** anında devre kararlı haldedir. (Vc(0)=2.5V , iL(0)= -1A)

21.12.2016

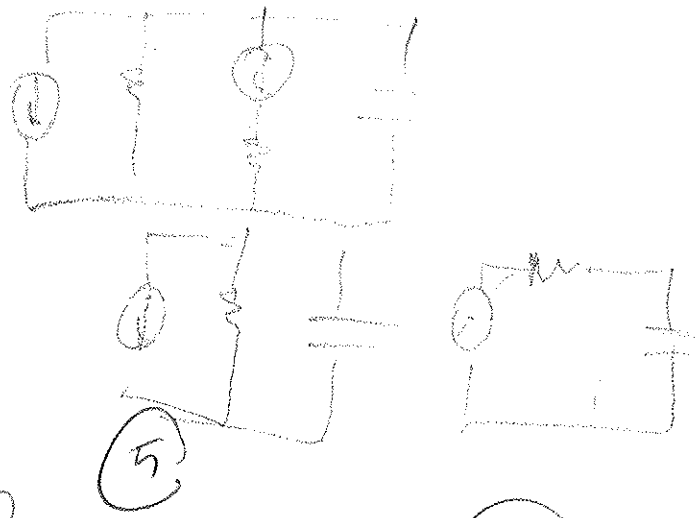
ELM 221 Devre Teorisi Hızlı Sınav 4

Adı-Soyadı:

Numara:



SORU: Şekildeki devrede $Vc(0^-) = 3$ V ise t ≥0 için Vc(t) nedir?



$$\frac{V_c - 12}{4.10^3} + 2.10^{-3} + C\frac{dV_c}{dt} = 0$$

$$\frac{V_c}{4.10^3} - \frac{12.10^{-3}}{4} + 2.10^{-3} + 10^{-6}\frac{dV_c}{dt} = 0$$

$$\frac{dV_c}{dt} + \frac{10^3}{4}V_c = 10^{-3} \Rightarrow V_{ch} \rightarrow s + 250 = 0 \quad s = -250$$

$$V_{ch} = A.e^{-250t}$$

$$V_{cp} = K \qquad V_{cp}' = 0$$

$$0 + 250K = 1000$$

$$K = 4$$

$$V_c = 4 + A.e^{-250t}$$

$$t=0 \rightarrow 4 + A = 3 \Rightarrow A = -1$$

$$V_c(t) = 4 - e^{-250t}$$

5

2,5

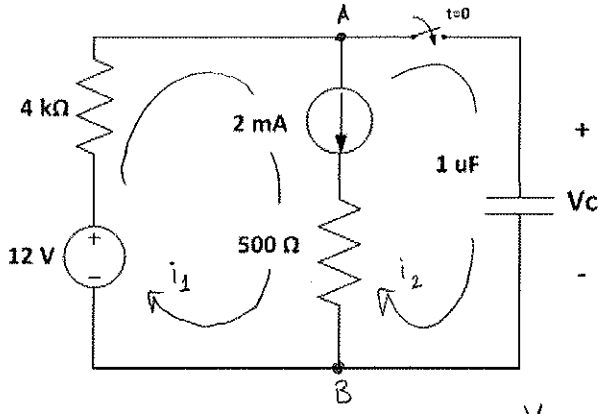
2,5

21.12.2016

ELM 221 Devre Teorisi Hızlı Sınav 4

Adı-Soyadı: Mustafa Asım YAYLAKAŞI

Numara: 141024012

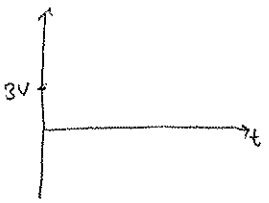


SORU: Şekildeki devrede $Vc(0^-) = 3$ V ise t ≥0 için Vc(t) nedir?

$t < 0$ 3V

$t \geq 0$?

$i_1 = 2\,mA$ $V_C = V_{AB}$



formüller

$i_c = C\frac{dV}{dt}$	$V = L\frac{di}{dt}$
$E_1 = \frac{1}{2}CV^2$	$\frac{1}{2}LI^2$
$\frac{1}{C}\int_0^t V\,dt$	$\frac{1}{L}\int_0^t I\,dt$

integral alan

Türev eğim

Çevre 2

$+V_C + 500(i_2 - 2) = 0$

$500 i_2 - 1000 = -V_C$

$500 i_2 = 1000 - V_C$

→ Kondansatörün yükü birden değişmeyeceği için

$500 i_2 = 997$

$i_2 = \frac{997}{500}\,mA \Rightarrow \frac{997}{500}\,mA = 1uF \cdot \frac{dV}{dt} \Rightarrow \frac{997}{500} \cdot \frac{1}{1000} \cdot \frac{1000000}{1} \Rightarrow V_c(t) = 1,994V$

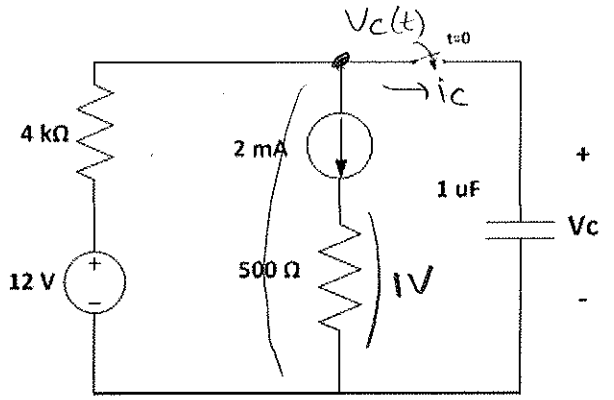
21.12.2016

ELM 221 Devre Teorisi Hızlı Sınav 4

10

Adı-Soyadı: Osman Yüksel

Numara: 121024001



SORU: Şekildeki devrede $Vc(0^-) = 3$ V ise t ≥0 için Vc(t) nedir?

$$\frac{Vc(t)-12}{4000} + 2\times10^{-3} + C\cdot\frac{dVc(t)}{dt} = 0 \qquad Vc(t)=y$$

$$1\times10^{-6}\cdot y' - \frac{12}{4000} + 2\times10^{-3} + \frac{y}{4000} = 0$$

$$y'(1\times10^{-6}) + y\left(\frac{1}{4000}\right) = 10^{-3}$$

Homojen Çözüm (Natural Response)

$$\alpha(1\times10^{-6}) + \frac{1}{4000} = 0$$

$$\alpha = -250 \quad \checkmark$$

$$y_H = C_1\cdot e^{-250t}$$

Özel çözüm (Forced Response)

$$y_p = B$$

$$B\cdot\frac{1}{4000} = 10^{-3} \quad \checkmark$$

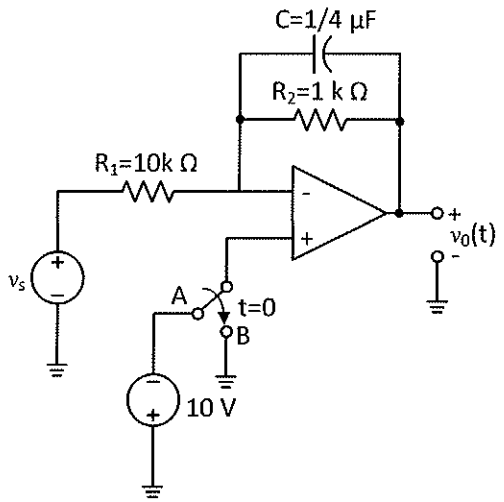
$$B = 4$$

21.12.2016

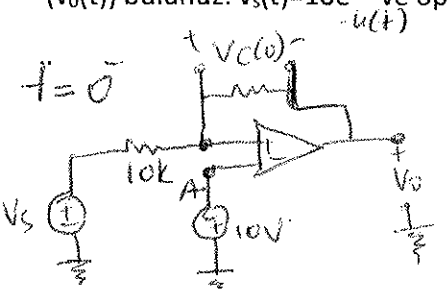
ELM 221 Devre Teorisi Hızlı Sınav 3

Adı-Soyadı:

Numara:

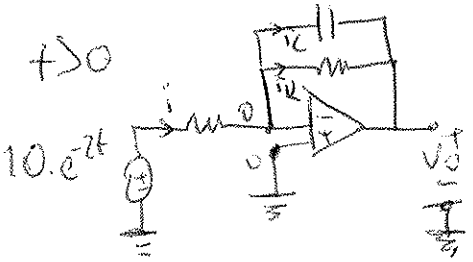


SORU: Şekildeki devrede t=0 anından önce anahtar uzun bir süre A konumunda kalmaktadır. t=0 anında anahtar A konumundan B konumuna geçmektedir. Çıkış gerilimi v_0 geriliminin zamana bağlı ifadesini ($v_0(t)$) bulunuz. $v_s(t)=10e^{-2t}$ ve opamp idealdir.



$V_A = -10V$

$V_S = 0$



$V_o = -V_c$

$\frac{-10-0}{10k} + \frac{-10-V_o}{1k} = 0 \Rightarrow V_o = -11V$

$V_c(0^-) = V_a - V_o = 1V$ (2)

$i = i_R + i_C$

$i = \frac{10.e^{-2t}}{10k} = 0,001e^{-2t}$

$i = i_R + i_C \Rightarrow 0,001.e^{-2t} = \frac{C dV_c}{dt} + \frac{V_c}{1000} \Rightarrow \frac{dV_c}{dt} + 4000\,V_c = 4000.e^{-2t}$ (4)

($C = \frac{1}{4} \cdot 10^{-6}$)

$s + 4000 = 0 \Rightarrow s = -4000$

$V_{c_h} = C_0 e^{-4000t}$ $\quad V_{cp} = Ae^{-2t}$ $\quad V_{cp}' = -2Ae^{-2t}$

$V_{cp} = -2Ae^{-2t} + 4000Ae^{-2t} = 4000e^{-2t}$ (2)

$A = \frac{4000}{3998} = 1,0005$

$V_c = C_0.e^{-4000t} + 1,0005\,e^{-2t}$

$V_c(0) = 1 = C_0 + 1,0005 \Rightarrow C_0 = -0,0005$ (2)

$V_c(t) = -0,0005\,e^{-4000t} + 1,0005\,e^{-2t}$

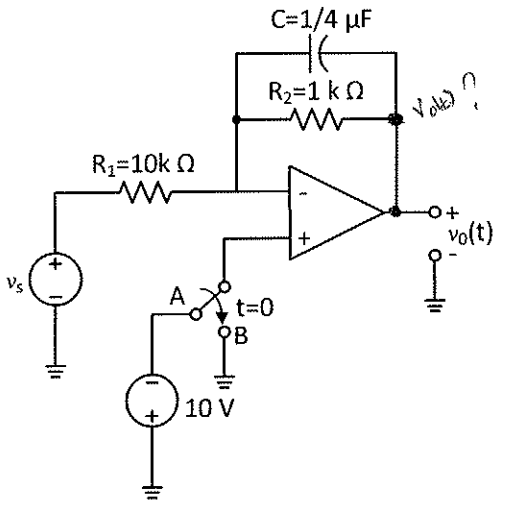
21.12.2016

10

ELM 221 Devre Teorisi Hızlı Sınav 3

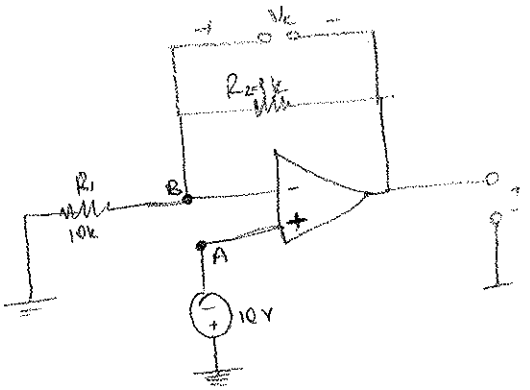
Adı-Soyadı: Burak Özdemir

Numara: 161024080



SORU: Şekildeki devrede t=0 anından önce anahtar uzun bir süre A konumunda kalmaktadır. t=0 anında anahtar A konumundan B konumuna geçmektedir. Çıkış gerilimi v_0 geriliminin zamana bağlı ifadesini ($v_0(t)$) bulunuz. $v_s(t)=10e^{-2t}$ ve opamp idealdir.

$t < 0$



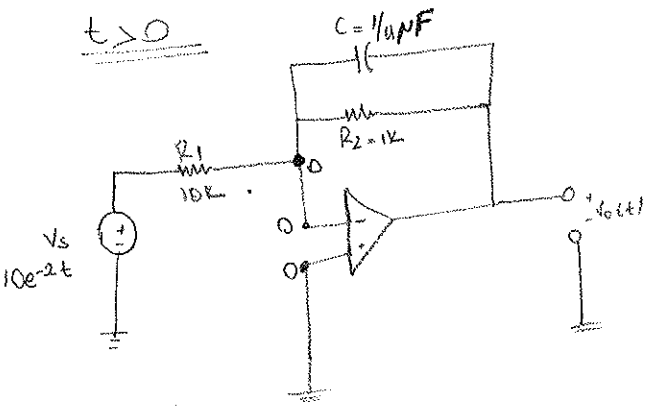
$V_B = -10V$

$$\frac{-10}{10k} = -\frac{V_c(0)}{1k}$$

$$V_c(0) = 1V$$

2

$t > 0$



$$\frac{V_c}{R_2} + C\frac{dV_c}{dt} + \frac{0-V_s}{R_1} = 0$$

$d/dt = s$

$$= \frac{V_c}{C\cdot R_2} + \frac{dV_c}{dt} = \frac{V_s}{C\cdot R_1}$$

$$= V_c\left(\frac{1}{C R_2} + s\right) = \frac{V_s}{C\cdot R_1} \Rightarrow V_c\left(\frac{4\cdot 10^3}{10^{-3}} + s\right) = \frac{4\times 10^6}{10\cdot 10^3}\cdot 10e^{-2t}$$

$$= V_c(4000 + s) = 4000e^{-2t}$$

4

$$= 4000\,V_c + \frac{dV_c}{dt} = 4000e^{-2t}$$

Homojen

$$= \frac{dV_c}{dt} + 4000V_c = 0$$

$$V_c(s+4000) = 0$$

$$s = -4000$$

$$V_{ch}(t) = A.e^{-4000t}$$

$$V_{cf}(t) = K.e^{-2t} \Rightarrow V_{cf}'(t) = -2Ke^{-2t}$$

$$= -2Ke^{-2t} + 4000K.e^{-2t} = 4000e^{-2t}$$

$$3998Ke^{-2t} = 4000e^{-2t}$$

$$K = \frac{4000}{3999} = 1,0005$$

$$V_c(t) = V_{ch} + V_{cf} = A.e^{-4000t} + 1,0005 = V_c(0)$$

$$A + 1,0005 = 1$$

$$A = -0,0005$$

2

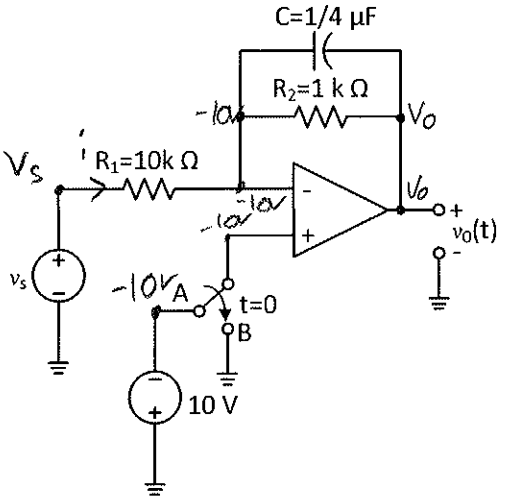
21.12.2016

ELM 221 Devre Teorisi Hızlı Sınav 3

Adı-Soyadı: Fatih Burak Pehlivan

Numara: 131026067

0



SORU: Şekildeki devrede t=0 anından önce anahtar uzun bir süre A konumunda kalmaktadır. t=0 anında anahtar A konumundan B konumuna geçmektedir. Çıkış gerilimi v_0 geriliminin zamana bağlı ifadesini ($v_0(t)$) bulunuz. $v_s(t)=10e^{-2t}$ ve opamp idealdir.

t<0 iken

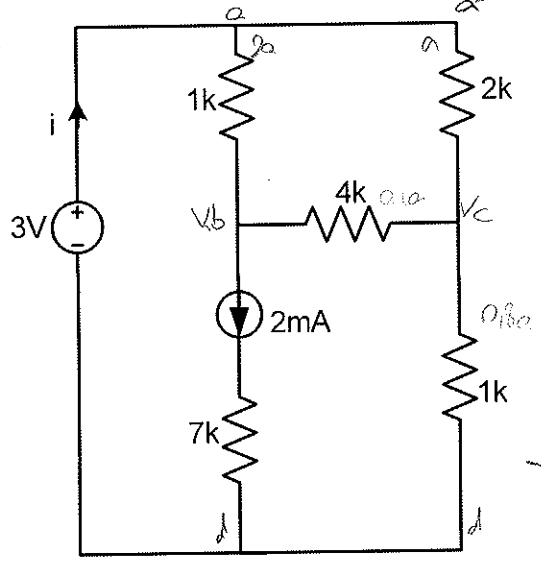
$\frac{V_s-(-10)}{10k\Omega}=i$ $\quad$ $\frac{-10-V_o}{1k\Omega}=i$

10.11.2016

ELM 221 Devre Teorisi Hızlı Sınav 2

Adı-Soyadı: ERCAN SAVAŞ

Numara: 17102049



SORU: Şekildeki devrede gösterilen i akımını mesh analiz kullanarak bulunuz.

Vb

$$\frac{V_b - V_c}{4k} + \frac{V_b - 3V}{7k} + \frac{3V - V_b}{1k} + 2mA = 0$$

$$7V_b - 7V_c + 4V_b - 12V + 8V - 28V_b + 56mA = 0$$

$$72V + 56mA - 17V_b - 7V_c = 0$$

$$17V_b + 7V_c = -72V - 56mA$$

Vc

$$\frac{V_c - V_b}{4k} + \frac{3V - V_c}{2k} + \frac{V_c - 3V}{1k} = 0$$

$$V_c - V_b + 6V - 2V_c + 4V_c - 12V = 0$$

$$V_c - V_b - 6V = 0$$

$$V_c - V_b = 6V$$

$$V_c - 4{,}4 = 6V$$

$$V_c = 10{,}4$$

$$\frac{V_b - 3V}{7k} = 2mA$$

$$V_b - 3V = 14V$$

$$V_b = 4{,}4V$$

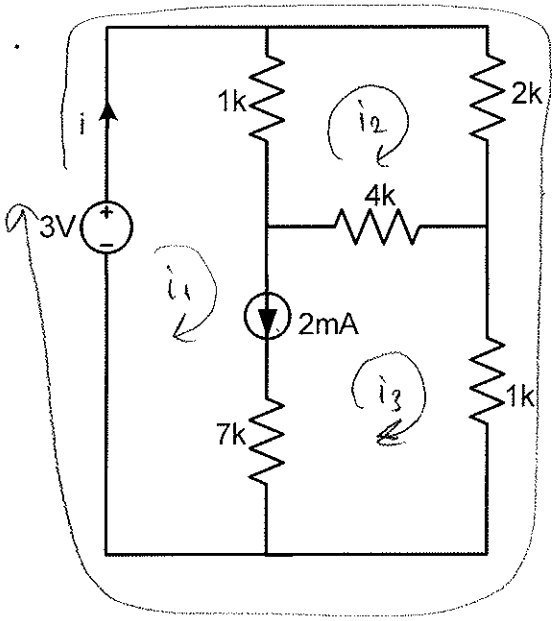
10.11.2016

ELM 221 Devre Teorisi Hızlı Sınav 2

10

Adı-Soyadı: Fatma Nur Yılmaz

Numara: 131024005



SORU: Şekildeki devrede gösterilen i akımını mesh analiz kullanarak bulunuz.

$3 = 2k\,i_2 + 1k\,i_3$

$2k\,i_2 + 4k(i_2 - i_3) + 1k\,(i_2 - i_1) = 0$ (3)

$i_1 - i_3 = 2 \Rightarrow i_1 = i_3 + 2mA$ (3)

$\Rightarrow 7k\,i_2 - 4k\,i_3 - k\,(i_3 + 2mA) = 0$

$7k\,i_2 - 5k\,i_3 = 2 \Rightarrow 14k\,i_2 = 14$

$5 / \; 2k\,i_2 + k\,i_3 = 3$

$i_2 = 1mA$

$5k\,i_3 = 7 - 2 = 5$

$\Rightarrow i_3 = 1mA$ ✓

$i_1 - i_3 = 2mA \Rightarrow i_1 - 1 = 2 \Rightarrow i_1 = 3mA$

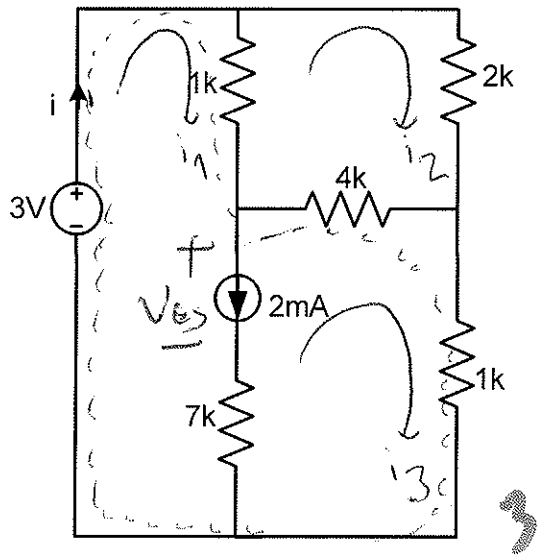
$i_1 = i = 3mA$ ✓

10.11.2016

ELM 221 Devre Teorisi Hızlı Sınav 2

Adı-Soyadı:

Numara:



SORU: Şekildeki devrede gösterilen i akımını mesh analiz kullanarak bulunuz.

Çevre 1 (Süpermesh)

$$-3+(i_1 \cdot 1k - i_2 \cdot 1k)+(i_3 \cdot 4k - i_2 4k)+i_3 \cdot 1k = 0$$

$$i_1 \cdot 1k - i_2 5k + i_3 5k = 3$$

3 $\boxed{i_1 - 5i_2 + 5i_3 = 3mA}$ (1)

Çevre 2

3 $(i_2 - i_1)1k + i_2 2k + (i_2 - i_3)4k = 0$

$$-1k i_1 + 7k i_2 - 4k i_3 = 0$$

$\boxed{-i_1 + 7i_2 - 4i_3 = 0}$ (2)

3 Akım kaynağı

$\boxed{i_1 - i_3 = 2mA}$ (3)

$i_1 = 3mA$
$i_2 = 1mA$
$i_3 = 1mA$

1

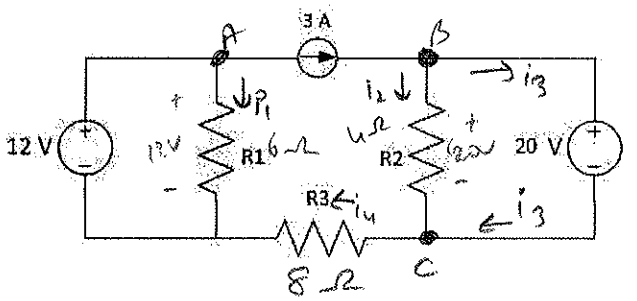
18.10.2016

ELM 221 Devre Teorisi Hızlı Sınav 1

10

Adı-Soyadı: Yusuf Can Sahin

Numara: 141024009



SORU: Şekildeki devrede ideal ve özdeş 2 gerilim kaynağı ve ideal 1 akım kaynağı bulunmaktadır. Şekildeki dirençlerin değerleri, R1=6Ω, R2=4Ω ve R3=8Ω'dur. Gerilim kaynakları ise 12 V ve 20 V potansiyeldedir. Akım kaynağı devreye soruda verildiği yönde 3 A akım sağlamaktadır. Buna göre :

a) Devredeki her bir direncin üzerinde harcanan gücü bulunuz.
b) Her bir kaynağın devreye sağladığı gücü bulunuz.
c) a ve b şıklarında bulduğunuz güçleri karşılaştırınız

Kirchoff

→ R_1, 12 V'luk gerilim kaynağına paralel

→ R_2, 20 V'luk gerilim kaynağına paralel

$P = I^2 R$
$P = V \cdot I$

$I_1 = \frac{12V}{R_1} = \frac{12V}{6\Omega} = 2A$

$I_2 = \frac{20V}{R_2} = \frac{20V}{4\Omega} = 5A$

a)

R_1	R_2	R_3
$P_1 = I_1^2 \cdot R_1$	$P_2 = I_2^2 \cdot R_2$	$P_3 = I_4^2 \cdot R_3$
$P_1 = 2^2 \cdot 6$	$P_2 = 5^2 \cdot 4$	$P_3 = 3^2 \cdot 8$
$P_1 = 24W$ ✓	$P_2 = 100\ W$ ✓	$P_3 = 72W$ ✓

noktasına göre KLC yapılırsa,

→ $-3A + I_2 + I_3 = 0$

→ $-3A + 5A + I_3 = 0$

$I_3 = -2A$

b)

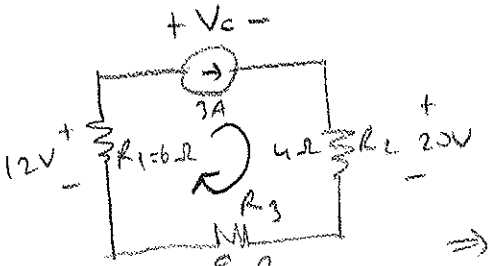
12V	20V
$P = V \cdot I$	$P = V \cdot I$
$P = (+12)(5) = -60W$ ✓	$P = 20 \cdot (-2) = -40W$ ✓

noktasına göre KLC yapılırsa

→ $I_4 - I_2 - I_3 = 0$

→ $I_4 - 5A + 2A = 0$

$I_4 = 3A$



$V = i \cdot R$
$V = 8 \cdot 3$
$V = 24V$

$\Rightarrow -12 + V_c + 20 + 24 = 0$

$V_c = 12 - 44$

$V_c = -32\ V$ ✓

$P = V \cdot i$
$P = (-32)(3) = -96\ W$

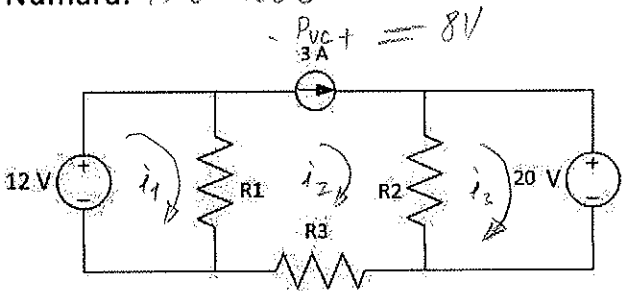
Arkada devam

18.10.2016

ELM 221 Devre Teorisi Hızlı Sınav 1

Adı-Soyadı: Furkan Karagöz

Numara: 151024006



SORU: Şekildeki devrede ideal ve özdeş 2 gerilim kaynağı ve ideal 1 akım kaynağı bulunmaktadır. Şekildeki dirençlerin değerleri, R1=6Ω, R2=4Ω ve R3=8Ω'dur. Gerilim kaynakları ise 12 V ve 20 V potansiyeldedir. Akım kaynağı devreye soruda verildiği yönde 3 A akım sağlamaktadır. Buna göre :

a) Devredeki her bir direncin üzerinde harcanan gücü bulunuz.

b) Her bir kaynağın devreye sağladığı gücü bulunuz.

c) a ve b şıklarında bulduğunuz güçleri karşılaştırınız

$P_{VC+} = 8V$

a) $P = V \cdot i$

$P_{VC} = 20 - 12 = 8$

$(i_1 - i_2)R_1 - 12 = 0$

$(i_2 - i_3)R_2 + i_2 R_3 + (i_2 - i_1)R_1 - 8 = 0$

$(i_3 - i_2)R_2 + 20 = 0$

$R_1 = 6$, $R_2 = 4$, $R_3 = 8$

$i_1(R_1) + i_2(-R_1) = 12$

$i_1(-R_1) + i_2(R_2 + R_3 + R_1) + i_3(-R_2) = 8$

$i_2(-R_2) + i_3(R_2) = -20$

$$i_1 = \frac{\begin{vmatrix} 12 & 6 & 0 \\ 8 & 18 & -4 \\ 20 & -4 & 4 \end{vmatrix}}{\begin{vmatrix} 6 & -6 & 0 \\ -6 & 18 & -4 \\ 0 & -4 & 4 \end{vmatrix}} = \qquad i_2 = \frac{\begin{vmatrix} 6 & 12 & 0 \\ -6 & 8 & -4 \\ 0 & 20 & 4 \end{vmatrix}}{\Delta} = \qquad i_3 = \frac{\begin{vmatrix} 6 & 6 & 12 \\ -6 & 18 & 8 \\ 0 & -4 & 20 \end{vmatrix}}{\Delta} =$$

b) $P_{12V} = 12V \cdot i_1$

$P_{20V} = 20V \cdot i_2$

$P_{8A} = 8V \cdot i_3$ watt güç sağlar

c) a şıkkında bulunan güç toplamları ile b şıkkında bulunan güç toplamlarının aynı olması beklenir.

(Akımları sayısal hesaplamak için yeterli vakit verilmedi.)

$\Rightarrow P_1 = 6 i_1^2$

$P_2 = 4 i_2^2$

$P_3 = 8 i_3^2$

CIRCUIT THEORY - I MAKE-UP EXAMINATION 26/01/2017

Q 1	Q 2	Q 3	Q 4	TOTAL

Name Surname :

School Number : MTT

Exam. Duration is 120 min. All questions have equal point.

Congratulations.

Q 1:

For the circuit below **R1=6kΩ, R2=4kΩ, R3=2kΩ, R4=1kΩ .**:

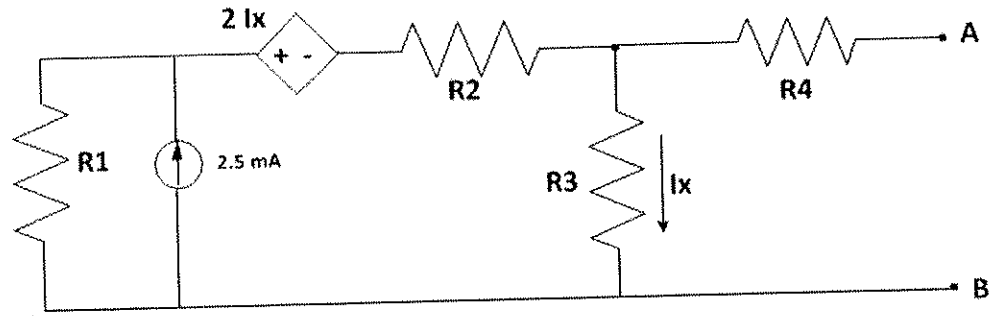
a) Find the **Thevénin Equivalent Circuit** between A-B nodes.

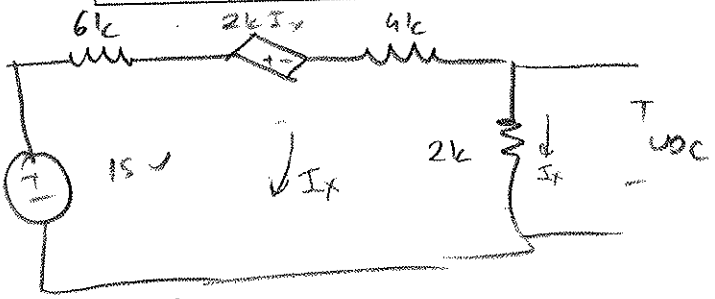
b) If the **maximum power transfer condition** is met for the circuit, what is the **resistor value**?

c) What is the **Maximum Power** transferred from source?

$V_{oc} = (6)$

$I_{sc} = (9)$

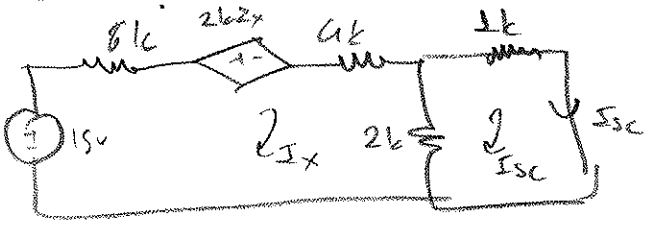




$-15V + 6kI_x + 6kI_x + 2kI_x = 0$

$\frac{15}{14} = I_x = 1{,}07\ mA$ (3)

$V_{oc} = 2kI_x = 2{,}14\ Volt$ (3)



$-15 + 6kI_x + 2kI_x + 4kI_x + 2kI_x - 2I_{sc} = 0$ (3)

$2I_{sc} - 2kI_x + I_{sc} = 0$ (3)

$I_x = \frac{3I_{sc}}{2}$

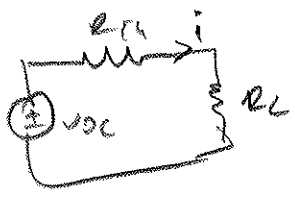
$-15 + 14I_x - 2I_{sc} = 0$

$I_{sc} = \frac{15}{13} = 0{,}78mA$ (3)

$R_{TH} = \frac{V_{oc}}{I_{sc}} = 2{,}74k$ (2)

b) $R_{Load} = R_{TH} = 2{,}74k\Omega$ (2)

c)



$i = \frac{V_{oc}}{2R_{TH}}$ $\quad P_L = i^2 R_L$

$P_L = \left(\frac{V_{oc}}{2R_{TH}}\right)^2 \cdot R_{TH} = \frac{V_{oc}^2}{4 \cdot R_{TH}} = \frac{4{,}57}{10{,}96} = 0{,}416\ mW$

1

mantık hatası var ise 1/4

işlem hatası ille hata (-3) şeklin yok hem de

mantık hatası bir yerde var

ama sonucu doğru ise "yarı" puan verdim

bilim hatası gün ışığına

CIRCUIT THEORY - I MAKE-UP EXAMINATION 26/01/2017

Q 1	Q 2	Q 3	Q 4	TOTAL

Name Surname :

School Number : MTT

Exam. Duration is 120 min. All questions have equal point.

Congratulations.

Q 1:

For the circuit below **R1=6kΩ, R2=4kΩ, R3=2kΩ, R4=1kΩ .**:

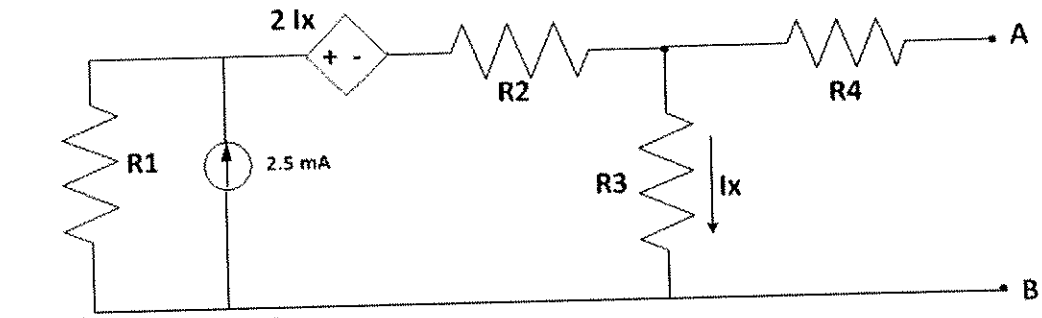
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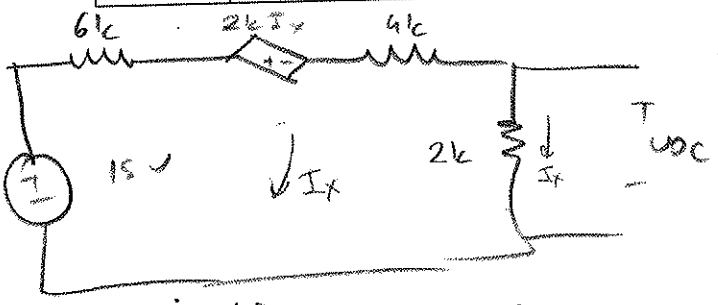
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$V_{oc} = (6)$

$I_{sc} = (9)$

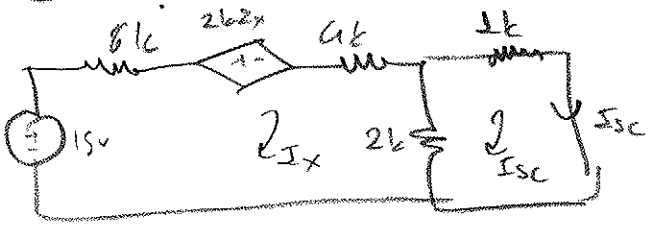




$-15V + 6kI_x + 6kI_x + 2kI_x = 0$

$\frac{15}{14} = I_x = 1{,}07\ mA$ (3)

$V_{oc} = 2kI_x = 2{,}14\ Volt$ (3)



$-15 + 6kI_x + 2kI_x + 4kI_x + 2kI_x - 2I_{sc} = 0$ (3)

$2I_{sc} - 2kI_x + I_{sc} = 0$ (3)

$I_x = \frac{3I_{sc}}{2}$

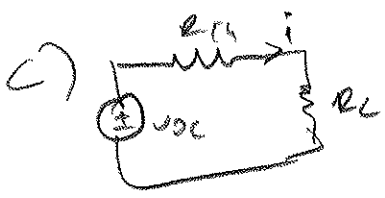
$-15 + 14I_x - 2I_{sc} = 0$

$I_{sc} = \frac{15}{13} = 0{,}78\ mA$ (3)

$R_{TH} = \frac{V_{oc}}{I_{sc}} = 2{,}74k$ (2)

b) $R_{Load} = R_{th} = 2{,}74k\Omega$ (2)

c)



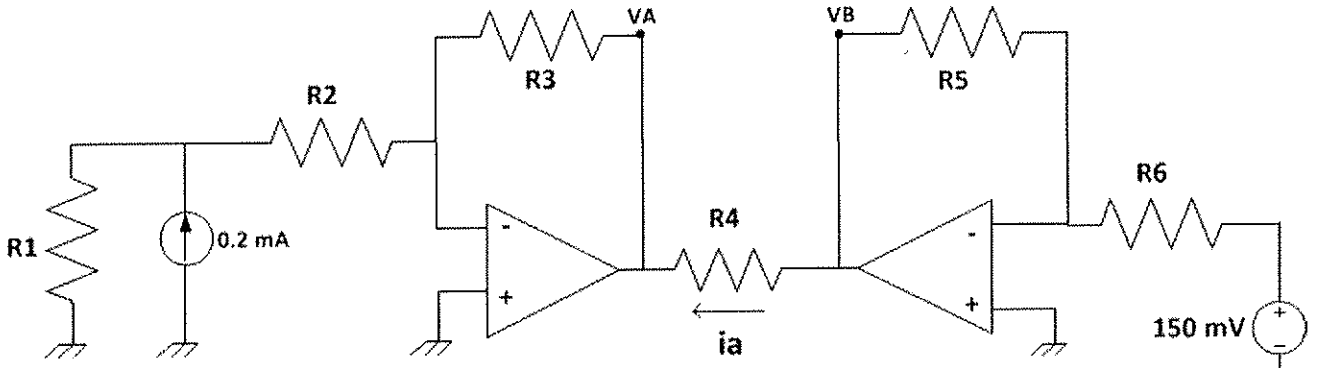
$i = \frac{V_{oc}}{2R_{th}}$ $P_L = i^2 \cdot R_L$

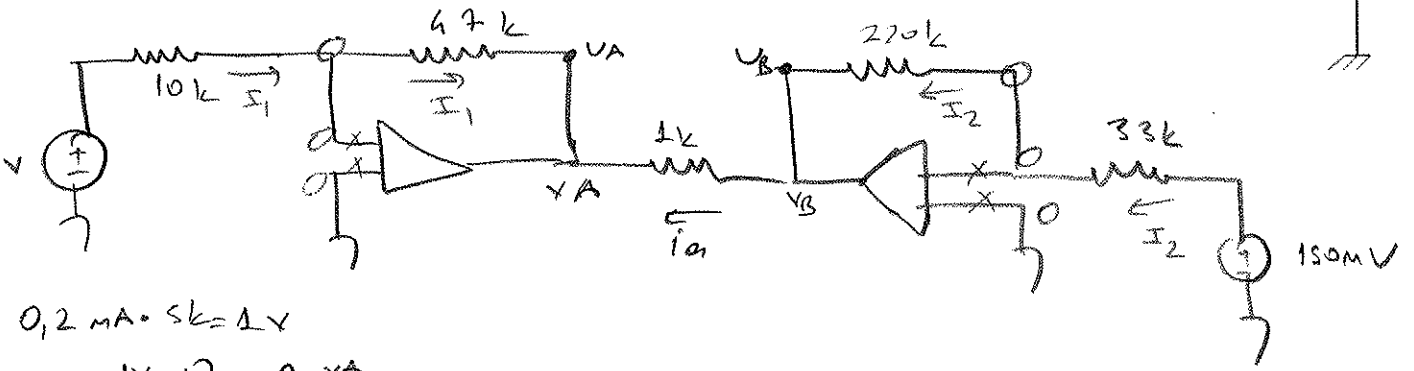
$P_L = \left(\frac{V_{oc}}{2R_{th}}\right)^2 \cdot R_{TH} = \frac{V_{oc}^2}{4 \cdot R_{TH}} = \frac{4{,}57}{10{,}36} = 0{,}416\ mW$ (4)

Ad Soyadı :

No :

Q 2: For the circuit below, **R1=5kΩ, R2=5kΩ, R3=47kΩ, R4=1kΩ, R5=220kΩ, R6=33kΩ.** **Find** the value of **ia current** by given direction that pass through on resistor R4





$0,2\ mA \cdot 5k = 1V$

$I_1 = \frac{1V-0}{10k} = \frac{0-VA}{47k}$

$VA = -4,7\ volt$

(8)

$I_2 = \frac{0-VB}{220k} = \frac{150mV-0}{33k}$

$VB = -\frac{150 \cdot 10^{-3} \cdot 220}{33} = -1V$

(10)

$i_a = \frac{VB - VA}{1k}$

$i_a = \frac{-1+4,7}{1k}$

$i_a = 3,7\ mA$

(6)

W ısları toplam

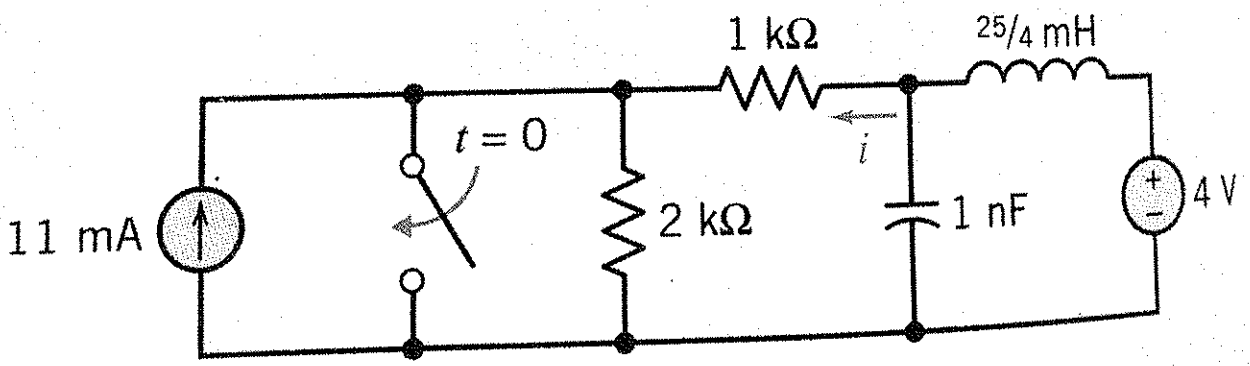
2 puan

Name Surname :

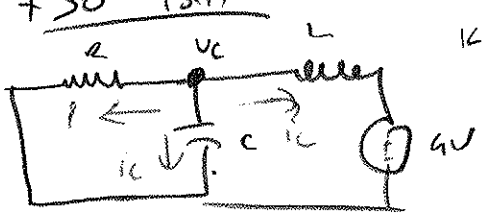
School Number :

Q3

P 9.9-1 Find $i(t)$ for $t > 0$ using the direct method for the circuit of Figure P 9.9-1.



$t>0$ için



KCL at Vc $\quad i_L + i_C + i = 0 \qquad i = \frac{V_c}{1000} \qquad i_C = C\frac{dv_c}{dt}$

türevini alırsak

$$\frac{di_L}{dt} + \frac{di_C}{dt} + \frac{di}{dt} = 0 \qquad i_L = \frac{1}{L}\int_{-\infty}^{t}(v_c - 4)\,dt$$

$$160(v_c - 4) + 10^{-9}\frac{dv_c^2}{dt} + 10^{-3}\frac{dv_c}{dt} = 0 \rightarrow \text{denkleği}$$

$$s^2 + 10^6 s + 1{,}6\cdot 10^{11} = 0 \quad (2)$$

$$\frac{dv_c^2}{dt} + 10^6\frac{dv_c}{dt} + 1{,}6\cdot 10^{11} v_c = 6{,}4\cdot 10^{11} \quad (6)$$ ise

$s_1 = -2\cdot 10^5$ (1)

$s_2 = -8\cdot 10^5$ (1)

$$v_{cn}(t) = A_1 \cdot e^{-2\cdot 10^5 t} + A_2 \cdot e^{-8\cdot 10^5 t} \quad (3)$$

$v_f = B$ ise bu ana denklemde yerine

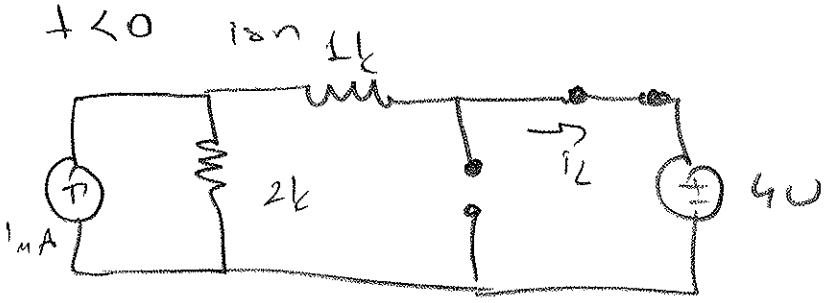
$0 + 0 + 1{,}6\cdot 10^{11} v_f = 6{,}4\cdot 10^{11}$

$v_f = B = 4$ (2)

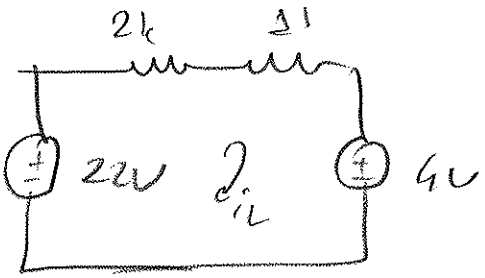
Tam cevap ise

$v_c(t) = v_n(t) + B$ dir.

$t<0$ için



$V_C(0) = 4V$ (2)



$-22 + 3i_L + 4 = 0$

$i_L(0) = 6mA$ (2)

$i_C = C \cdot \frac{dv_C}{dt}$ ise $\quad i_C(0) = C \cdot \frac{dv_C(0)}{dt}$

$i + i_C + i_L = 0 \Rightarrow \underline{i_C(0)} = -i - i_L = -\frac{V_C(0)}{1k} - i_L(0) = -4mA - 6mA = \underline{-10mA}$

$\frac{-10 \cdot 10^{-3}}{10^{-9}} = \frac{dv_C(0)}{dt} = \underline{\underline{-10^7}}$ (2)

$V_C(0) = 4 = A_1 \cdot e^0 + A_2 \cdot e^0 + 4$

$A_1 + A_2 = 0$ (1)

$\frac{dv_C(0)}{dt}$ Ana denklem türevi alınırsa

$= -2 \cdot 10^5 \cdot A_1 - 8 \cdot 10^5 \cdot A_2 = -10^7$ 2. denklem

2. denklemden (2)

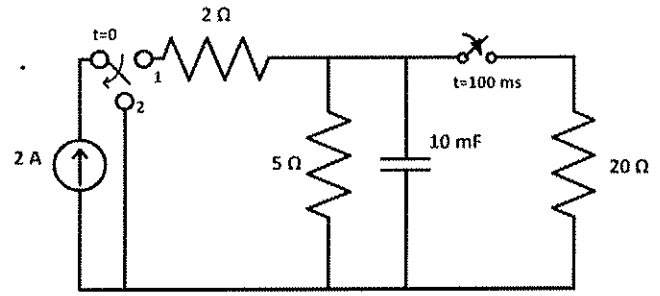
$A_1 = 50/3 \quad A_2 = -50/3$ yerine

$V_C(t) = \frac{50}{3} \cdot e^{-2 \cdot 10^5 t} - \frac{50}{3} e^{-8 \cdot 10^5 t} + 4$ (2)

$i(t) = \frac{V_C(t)}{1000} = \frac{50}{3000} \cdot e^{-2 \cdot 10^5 t} - \frac{50}{3000} e^{-8 \cdot 10^5 t} + 4 \cdot 10^{-3}$

Name Surname :

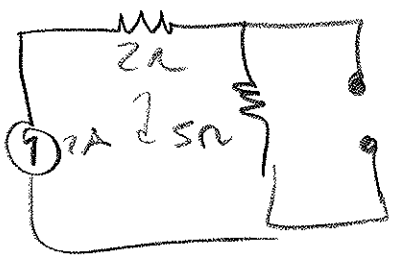
School Number :



Q 4:
Calculate the Vc(t) for t>0. Assume that the circuit is at steady-state condition at **t=0⁻**.

$V_{oc} + (V(0) - V_{oc}) \cdot e^{-t/\tau}$

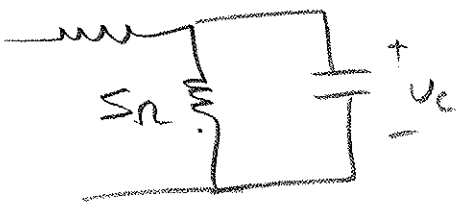
t<0



$5 \cdot 2 = 10V$

$V_C(0) = 10V$ (5)

0<t<100 ms



$V_C(\infty) = 0$ $R_T = 5\Omega$

$V_C = V_\infty + (V_0 - V_\infty) \cdot e^{-t/\tau}$

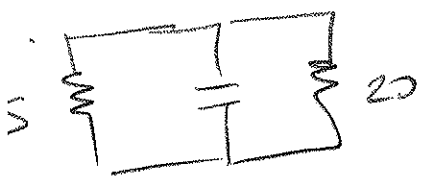
$\tau = C \cdot R_{TH} = 0{,}05$ (3)

$V_C = 0 + (10 - 0) \cdot e^{-t/0{,}05}$

$V_C = 10 \cdot e^{-20t}$ (7)

$V_C(100) = 10 \cdot e^{-2} = 1{,}35V$

100 < t



$\frac{20 \cdot 5}{20+5} = 4\Omega$

$R_{TH} = 4\Omega$

$V_C(\infty) = 0$ $V_C(0) = 10$

$\tau = C \cdot R_{TH} = 0{,}04$ (3)

$V_C(t) = V_\infty + (V_{(100ms)} - V_\infty) \cdot e^{-t/\tau}$

$V_C(t) = 1{,}35 \cdot e^{-25t}$ (7)

CIRCUIT THEORY - I MAKE-UP EXAMINATION 26/01/2017

Name Surname : Arda Coşkun

School Number : 141024028

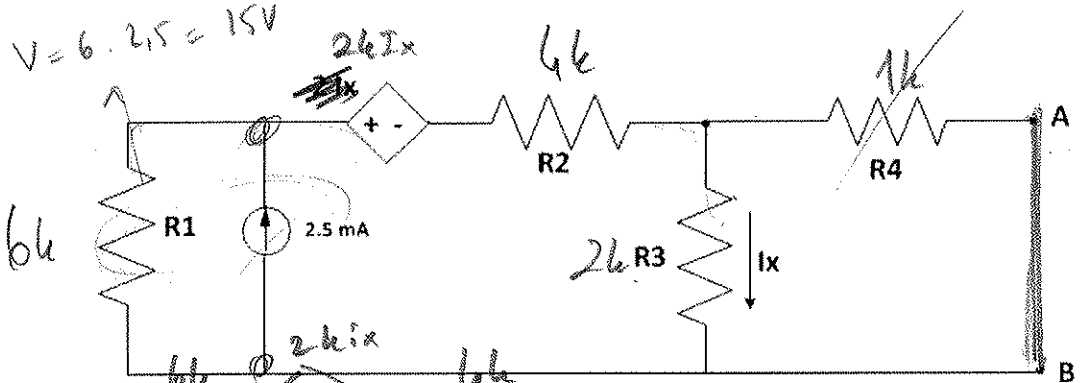
Q 1	Q 2	Q 3	Q 4	TOTAL

Exam. Duration is 120 min. All questions have equal point.
Congratulations.

Q 1:

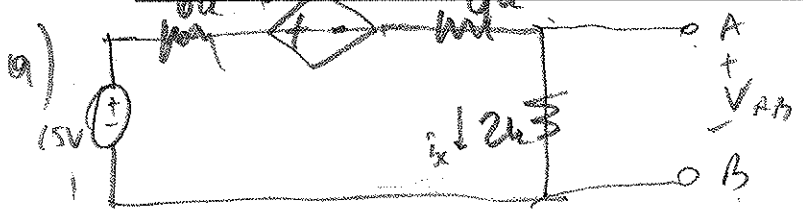
For the circuit below **R1=6kΩ, R2=4kΩ, R3=2kΩ, R4=1kΩ .**:

a) Find the **Thevénin Equivalent Circuit** between A-B nodes.
b) If the **maximum power transfer condition** is met for the circuit, what is the **resistor value**?
c) What is the **Maximum Power** transferred from source?



$V = 6 \cdot 2{,}5 = 15V$

a)



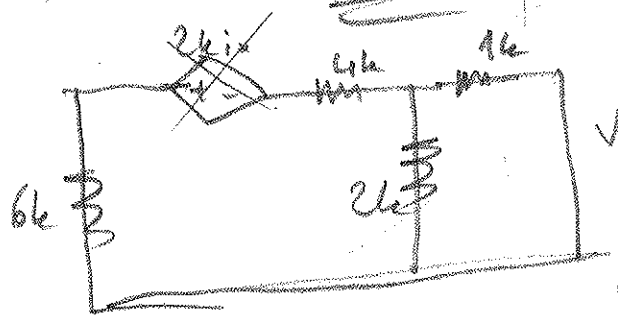
$V_{AB} = 2k \cdot i_x$

$-15V + 6i_x - 2i_x + 6i_x + 2i_x = 0$

$11i_x = 15$

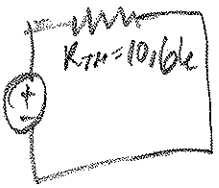
$i_x = 1{,}5\ A$

$V_{AB} = 2 \cdot 1{,}5 = 3V$



$V_{AB} = 0$ olur. Bunda dolayı $i_x = 0$ ve $2k i_x = 0$ olur.

$R_{eş} = \frac{32}{3} = 10{,}6k$



b) $R_{L,TH} = R_L$ olmalı.



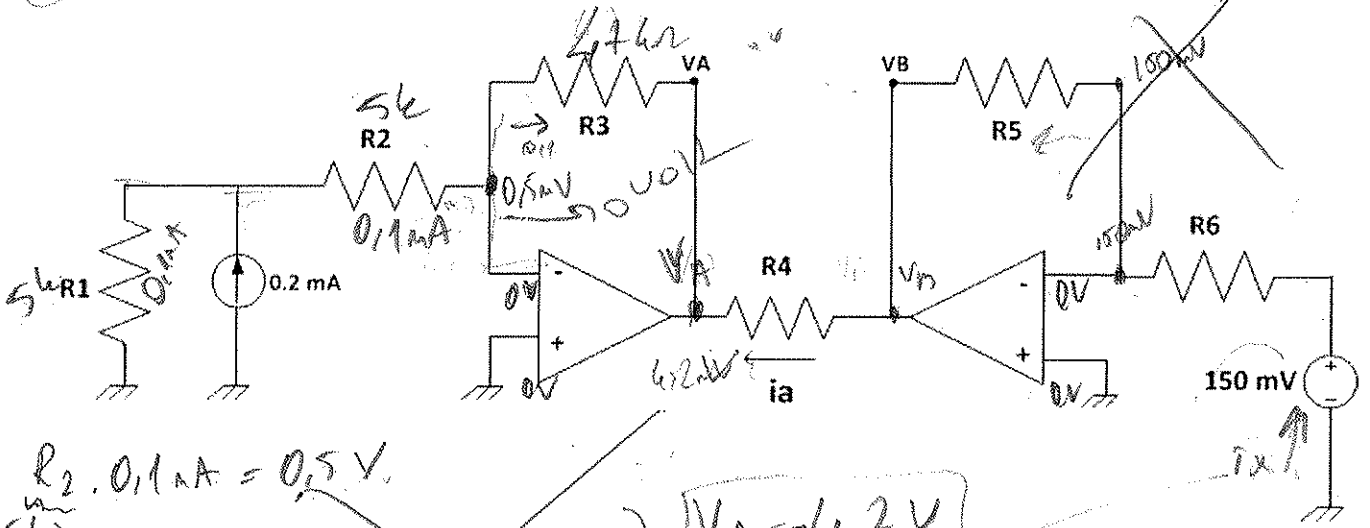
$R_L = 10{,}6$ olmalı.

c) max power $= i^2 \cdot R_{eş}$

$\left(\frac{3}{10{,}6}\right)^2 \cdot 10{,}6 = 0{,}301\ W$

Ad Soyadı : Arda Coşkun
No : 161024028

Q 2: For the circuit below, **R1=5kΩ, R2=5kΩ, R3=47kΩ, R4=1kΩ, R5=220kΩ, R6=33kΩ**
Find the value of **ia current** by given direction that pass through on resistor R4



$R_2 \cdot 0{,}1\,mA = 0{,}5\,V$

$\frac{0{,}5\,V - V_A}{47\,k\Omega} = 0{,}1\,mA \Rightarrow V_A = -4{,}2\,V$

$\left(\frac{V_B - V_A}{R_4}\right) = i_a$

$\frac{150\,mV}{R_6} = i_x$

$i_x = \frac{150\,mV}{33\,k\Omega} \simeq 4.55 \times 10^{-6}\,mA$

$V_B = -850\,mV$

$\left(\frac{150\,mV - V_B}{220\,k\Omega}\right) = 4.55 \times 10^{-6}\,mA$

$\left(\frac{V_B - V_A}{R_4}\right) = i_a$

$i_a = \frac{-850\,mV - 4{,}2\,V}{1\,k\Omega} = -5.050\,A$

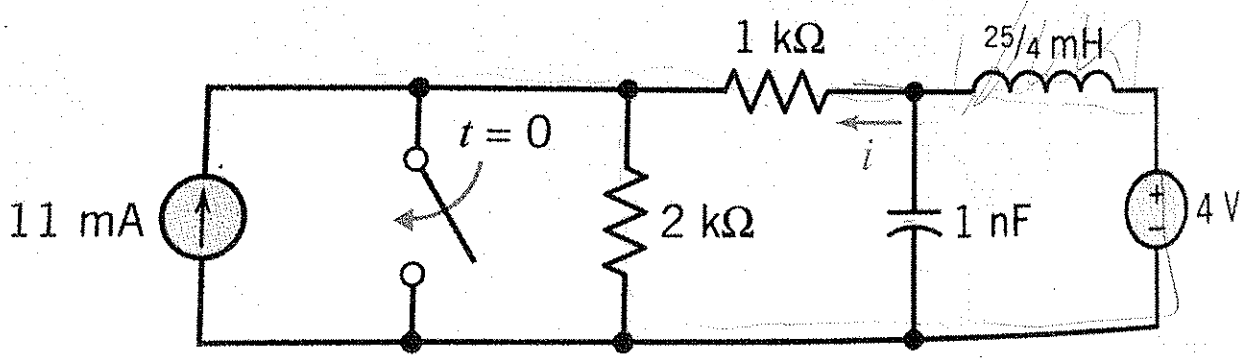
$i_a = -5{,}050\,A$

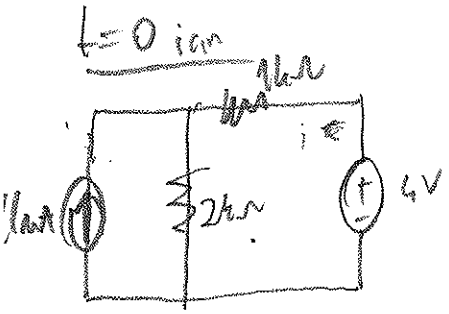
Name Surname : Arda Coşkun

School Number : 141024028

Q3

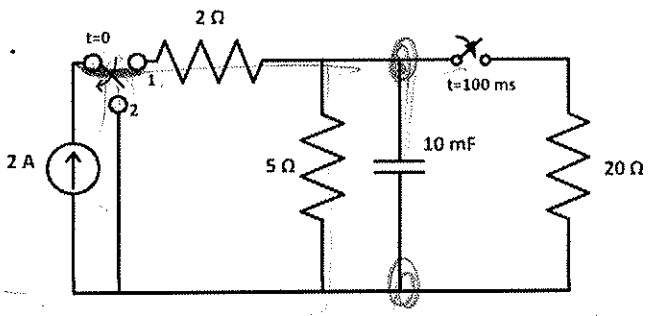
P 9.9-1 Find $i(t)$ for $t > 0$ using the direct method for the circuit of Figure P 9.9-1.





Name Surname : Arda Coşkun

School Number : 141024028

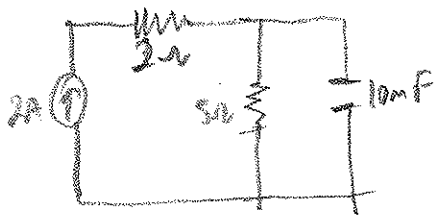


Q 4:
Calculate the Vc(t) for t>0. Assume that the circuit is at steady-state condition at **t=0⁻**. t=100 ms anahtar kapatılıyor.

t<0 için

$i(0^-) = i(0^+) = i(0) = 0A$

t=0 için



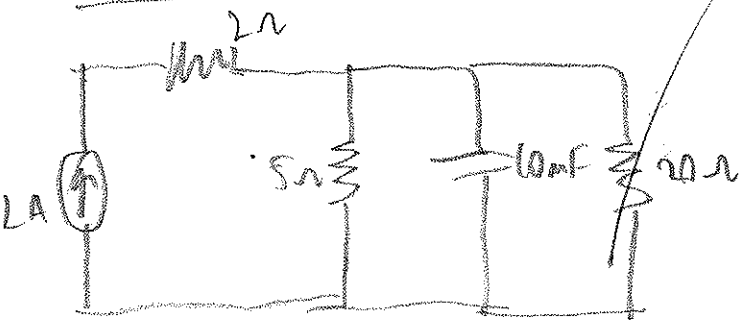
$\tau = \frac{L}{R} = \frac{10mF}{2.5} = 2ms$

$V(\infty) = V_0 \cdot e^{-t/\tau}$

$V_0 = \frac{20}{7}$

$V(\infty) = \frac{20}{7} \cdot e^{-t/2ms}$

t>0 için



CIRCUIT THEORY - I MAKE-UP EXAMINATION 26/01/2017

Name Surname : Mumme Taha Cece

School Number : 141024010

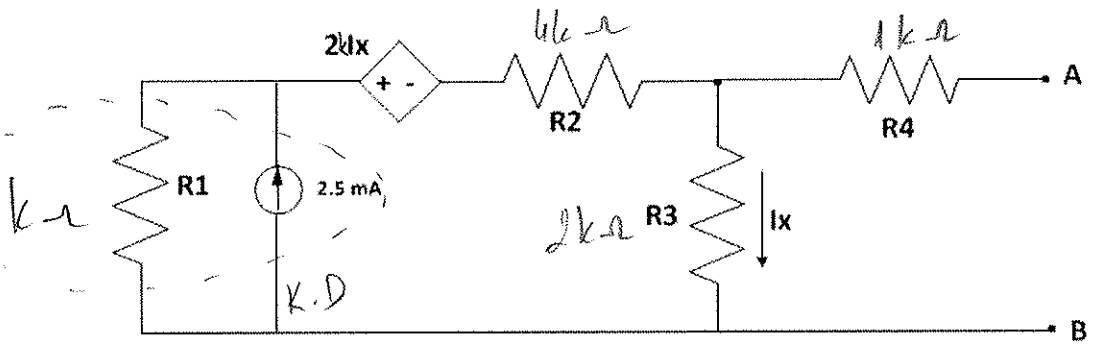
Q 1	Q 2	Q 3	Q 4	TOTAL
14	25	25	25	89

Exam. Duration is 120 min. All questions have equal point.
Congratulations.

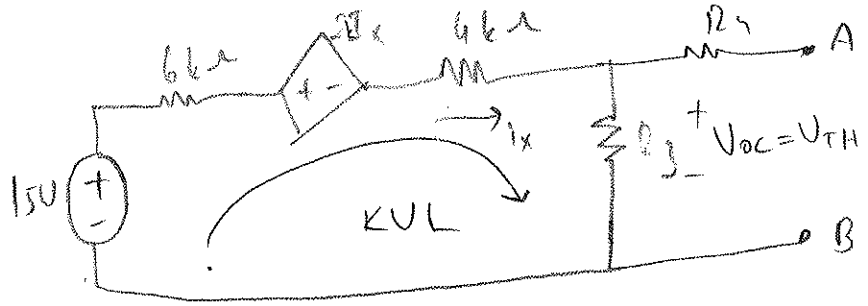
Q 1:

For the circuit below **R1=6kΩ, R2=4kΩ, R3=2kΩ, R4=1kΩ .**:

a) Find the **Thevénin Equivalent Circuit** between A-B nodes.
b) If the **maximum power transfer condition** is met for the circuit, what is the **resistor value**?
c) What is the **Maximum Power** transferred from source?



a)



Voc bulmak için

KVL

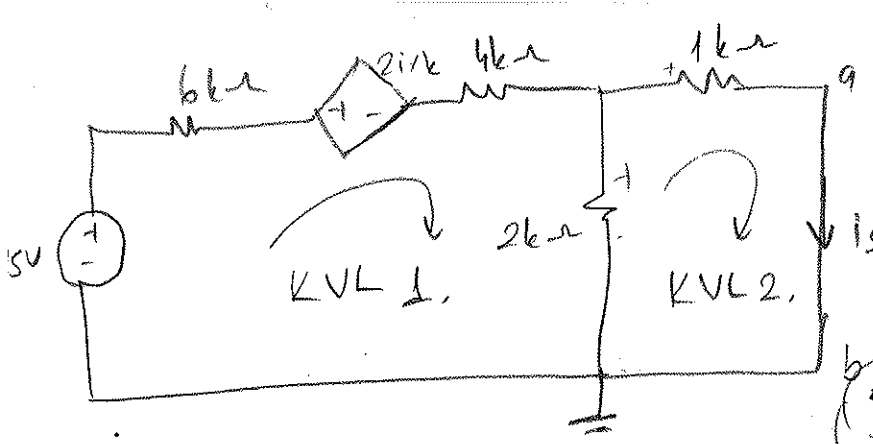
$-15 + 10ix.10^3 + 2.10^3 ix + 2.10^3 ix = 0$

$15 = 14 ix . 10^3$

$ix = \frac{15}{14}\ mA$ (3)

$R_3 . ix = V_{oc} \Rightarrow 2k\Omega . \frac{15}{14}\ mA$

$V_{oc} = \frac{15}{7}\ V$ (3)



KVL 1.

$-15 + 10.10^3 i_x + 2.10^3 i_x + 2.10^3 (i_x - i_{sc}) = 0$

$15 = 14.10^3 i_x - 2.10^3 i_{sc}$ -- (1)

KVL 2.

$10^3 - 2(i_{sc} - i_x) + 10^3 i_{sc} = 0$

$10^3 - i_{sc} + 2 i_x = 0$ -- (2). denklemden

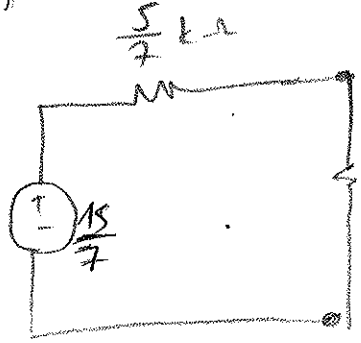
$i_{sc} = 2 i_x$ 1. KVL de yerine koy.

$15 = 14.10^3 i_{sc} - 2 i_{sc}.10^3$

$i_{sc} = 3\,mA$

$R_{TH} = \frac{V_{oc}}{I_{sc}} = \frac{\frac{15}{7}}{3\,mA} = \frac{5}{7}\,k\Omega$

b)



azami güç olması için $R_{TH} = R_L$ olması lazım

c) $P_{RL} = \frac{[V_{oc}]^2}{4 R_L} = \frac{\left[\frac{15}{7}\right]^2}{4 \cdot \frac{5}{7}.10^{3}} = \frac{4,6 \cdot .10^{-3}}{2,86}$

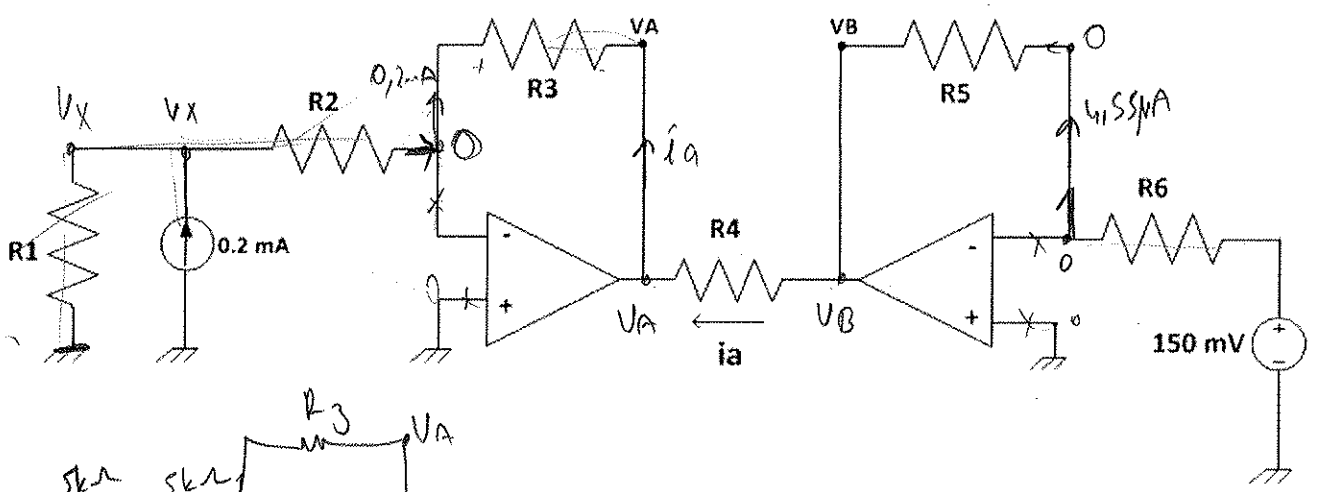
$P_{RL} = 1,61\,mW$

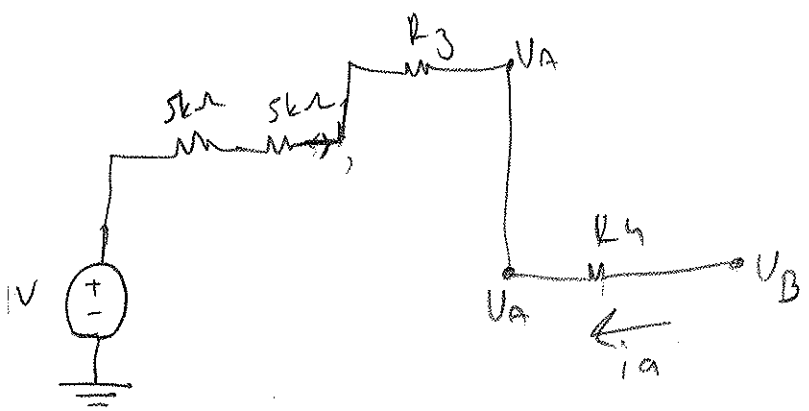
Ad Soyadı : Mumer Taha Ceren

No : 1402b010

Q 2: For the circuit below, **R1=5kΩ, R2=5kΩ, R3=47kΩ, R4=1kΩ, R5=220kΩ, R6=33kΩ.**
Find the value of **ia current** by given direction that pass through on resistor R4

$i_a = 3,7$





$\frac{150.10^{-3}}{33.10^{3}} = 4,55\,\mu A$

$\frac{1}{10k\Omega} = 0,1\,mA$

$4,55.10^{-6} \cdot R_5 = V_B$

$V_B = 4,55.220.10^{-6}.10^{3} = 1001\,V$

$0,1\,mA \cdot R_0 = V_A$

$V_A = 0,1.10^{-3}.47.10^{3}$

$V_A = 4,7\,V$

$\frac{V_B - V_A}{1k\Omega} = i_a$

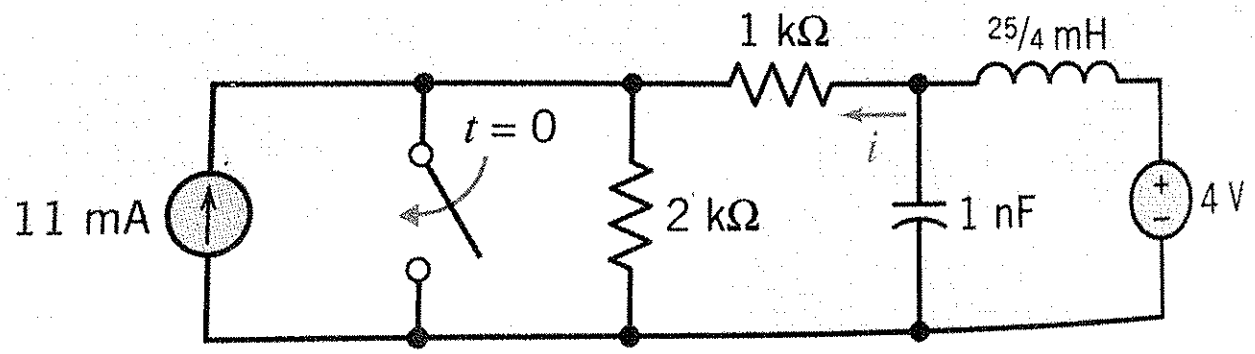
$\frac{1,001 - 4,7}{10^{3}} = i_a$

$i_a = -3,7\,mA$

Name Surname : Muammer Taha Ceren
School Number : 161026010

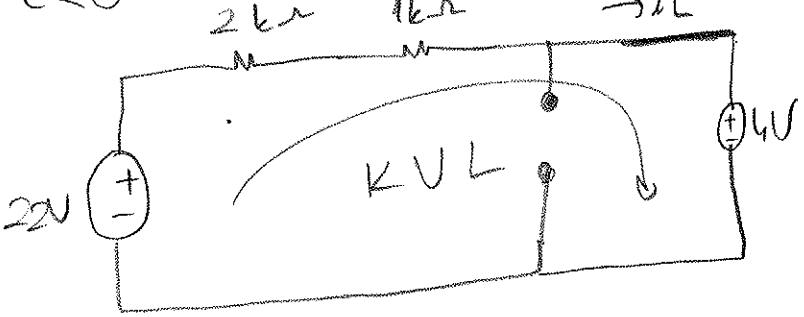
Q3

P 9.9-1 Find $i(t)$ for $t > 0$ using the direct method for the circuit of Figure P 9.9-1.



$U_C(0^-) = 4V$ ✓
$i_L(0^-) = 6mA$ ✓

$t<0$



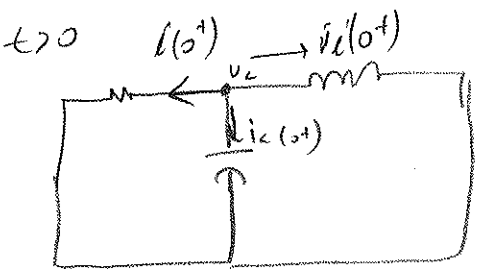
KVL

$-22V + 2k i_L + 1k i_L + 4V = 0$

$18V = 3k\, i_L$

$i_L = 6mA$

$t>0$



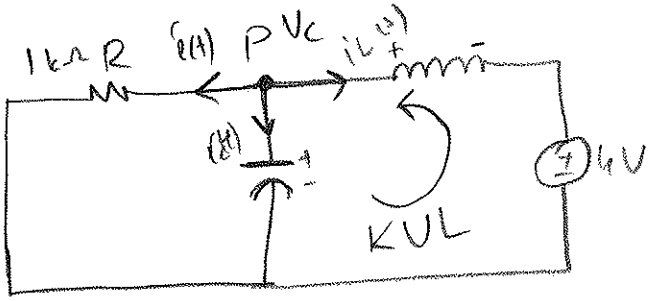
$\frac{V_c}{1k\Omega} + i_L(0^+) + i_c(0^+) =$

$\frac{V_c}{1k\Omega} = 4mA$, $i_L(0^+) = 6mA$

$i_c(0^+) = -10mA$ ✓

$C\frac{dV_c}{dt} = \frac{-10\cdot10^{-3}}{10^{-9}} = \frac{dV_c}{dt}$

$t > 0$



Node P Nok

Node P Nok.

$$\frac{V_c}{R} + i_c + i_L = 0$$

$$i_c = C \cdot \frac{dv_c}{dt}$$

$$\frac{V_c}{R} + \frac{dV_c \cdot C}{dt} + i_L = 0 \quad \ldots (1)$$

$$\frac{dV_c}{dt\,R\,C} + \frac{d^2V_c}{dt^2} + \frac{di_L}{dt\,C} = 0$$

$$\frac{d^2V_c}{dt^2} + \frac{dV_c}{dt\,RC} + \frac{V_c}{CL} = + \frac{V_s}{L}$$

$$s^2 + 10^6 s + 1.6.10^{11} = 6{,}4.10^{11}$$

$$V_{cn}(t) = s^2 + 10^6 s + 1{,}6.10^{11} = 0$$

s_1 ve s_2'yi nasıl bulduk?

$$V_{cn}(t) = A_1 \cdot e^{-8.10^5 t} + A_2 \cdot e^{-2.10^5 t}$$

$$V_c(t) = A_1 \cdot e^{-8.10^5 t} + A_2 \cdot e^{-2.10^5 t} + 4 = 0$$

$$V_c(0) = A_1 + A_2 + 4 = 4 \qquad \boxed{A_1 = -A_2}$$

$$\frac{dV_c(0)}{dt} = -8.10^5 \cdot A_1 \cdot e^{-8.10^5 t} - 2.10^5 \cdot A_2 \cdot e^{-2.10^5 t} = -10^7$$

$$-8.10^5 A_1 - 2.10^5 A_2 = -10^7$$

$$(+8A_1 + 2A_2) = +10^2$$

$$4A_1 + A_2 = 50$$

$$3A_1 = 50 \qquad \boxed{A_1 = \frac{50}{3} \quad A_2 = \frac{-50}{3}}$$

KVL

$$-V_s - V_L + V_c = 0$$

$$V_L = -V_s + V_c$$

$$\frac{di_L}{dt} = \frac{V_c - V_s}{L} \quad \ldots (2)$$

1. denklemin türevini alıp yerine yaz.

$$V_{cf}(t) = K$$

$$K'' + K' + 1{,}6.10^{11} = 6{,}4.10^{11}$$

$$K = 4$$

$$V_{cf}(t) = 4.$$

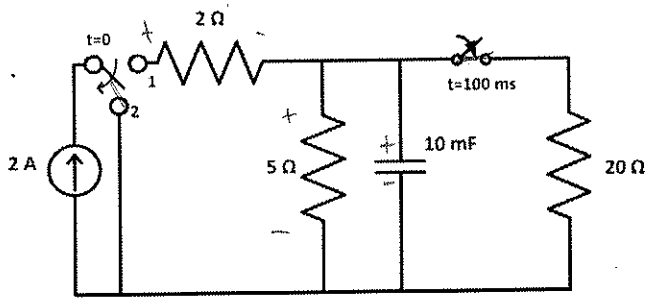
$$V_c(t) = \frac{50}{3} \cdot e^{-8.10^5 t} - \frac{50}{3} \cdot e^{-2.10^5 t} + 4$$

$$i(t) = \frac{V_c(t)}{1k\Omega} =$$

$$\boxed{i(t) = \frac{50}{0{,}1+3} \cdot e^{-8.10^5 t} - \frac{50}{2.1+3} \cdot e^{-2.10^{-5} t} + 4.10^{-3}}$$

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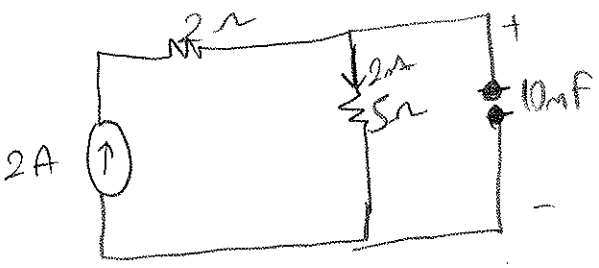
School Number : 141024010



Q 4:
Calculate the Vc(t) for t>0. Assume that the circuit is at steady-state condition at **t=0⁻**.

$\tau = R_{TH} \cdot C$

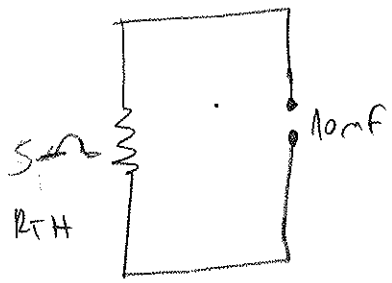
$t < 0$



$V_c(0^-) = 2A \cdot 5\Omega = 10V$

$t < 0 < 100ms \quad V(t) = V_{oc} + (V_0 - V_{oc}) \cdot e^{\frac{-t}{\tau}}$

$t < 0 < 100\,ms$



$V_{oc} = 0\,V$ (∞)

$V(t) = 10 \cdot e^{-20t}$

$\tau = 5 \cdot 10 \cdot 10^{-3}$

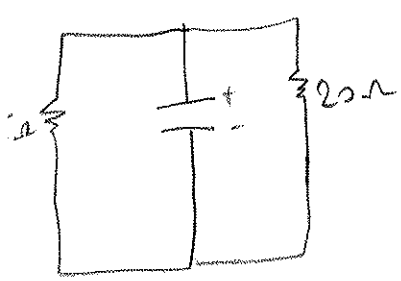
$\tau = 0{,}05\,sn$

$V_{0(t=0.1s)} = 10 \cdot e^{-20 \cdot 0{,}1s}$

$V_0 = 1{,}35\,V$

$t > 100\,ms \quad V(t) = 0V + (1{,}35 - 0) \cdot e^{-(t-0{,}1s)/\tau}$

$t > 100\,ms$



$V_{oc} = 0\,V$

$R_{TH} = 4\Omega$

$V(t) = 1{,}35 \cdot e^{-25(t-0{,}1sn)}$

$\tau = 4 \cdot 10 \cdot 10^{-3}$

$\tau = 0{,}04\,sn$